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本文将一个二维浅水方程（SWE）模型与便捷模块相结合，实现了对显示们发的提前潮汐扩展过程的模拟。无需预先设定潮汐的位置和尺寸作为输入条件。在用于模型验证的各项试验中，无论是针对制性还是非制性材料，模型均能准确预测潮汐的发展过程。

基于GPU加速的2D浅水方程模型中模拟潮汐扩展过程的高效模型
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研究目标与创新点
提出一种能够模拟由重顶引起的提前潮汐开启及后续流水过程的二维浅水方程（SWE）模型。
采用基于物理的便捷模型，可模拟由制性或非制性材料构成堤切的潮汐演化。
模型特点
模型基于GPU并行计算实现快速执行，适用于真实案例。
在设计时根据模型水动力条件对材料特性的依赖性，开展特殊假定潮汐位置和形状。
验证与应用
实现了CUDA加速程序，支持在图形处理器上运行，确保高分辨率网格和大域范围内的快速模拟。
基于多个实际测试进行验证，数值预测结果与实测数据趋势良好。
与基于Exner双相防沉积输运模型对比分析了性能表现，证明该模型对比计算效率的显著提升。
展示了一个历史关水事件的应用案例，表明该模型可用于真实场地模拟，能够反映潮汐演化。
背景介绍
低地地区通常依靠切向上流水保障，但切换操作可能导致堤切遭受开最终损害。
土地利用变化和气候变化可能增加特定堤切附下的负担，使现有情况不再足够。
动物洞穴等内部侵蚀过程也可能导致堤切失稳。
数值建模的重要性
堤切失效可能导致严重的经济和人员损失，因此设计流水风险预测和应急计划至关重要。
数值模拟是分析这些复杂现象的强大工具。
模拟方法比较
一维（1D）河道模型结合二维（2D）洪水区模型常常是现场最广泛使用的方法。
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潮汐形成过程具有三维性质，涉及复杂的水动力、解密精密性，需小幅动和表面过程交互作用。
实际应用中常采用大量简化方法，如忽视密切动态或精简为SWE与2D流动模型模型。
几何参数决定堤了材料特性，可能导致不准确预测。
参数化堤体模型主要针对三重考虑决定重开发，对侧向流流潮过通用性有限。
物理基础多维模型的优势
最佳选择是使用详细的物理基础多维模型。整合水动力和地质动力学方程。
可用时模拟切流，穿透口和淹没区域的流动，无需在潮口处引入内部边界条件。
应用于实际案例受限于堤切材料数据库信息和精确性以及计算时间成本的问题。
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细粒沉积输运模型，堤体建要通会描述潮口过程，因其包含特定可堤性参数是通用于制性堤切。
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二维深度平均模型通过充分模拟此三维过程，简单引入头切材料来建模此类潮汐。
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此操作在每时时增量每个单元（基于易修堤模型）的整前床面剪应力值修改面前推理。
实践中只有当堤切被重源时才会发生显著侵蚀。
可加性最低床面高程以避免引不下不可侵蚀堤的侵蚀。
仅勾床冲刷而不包括沉积，假设输侵蚀物质被水流带走。
保持两步大某以节省每大量，需调整这些参数从而在效率和精度之间取得平衡。
堤体过程依照下式确定（通过）和堤切材料（通过τ_{cd}，τ_{cr}）。
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尽管已有专门实验测试用于确定这些参数（如随参数强化型），但可靠的相关性尚未建立。
不同测试配置和结果解释会导致不同的k_d和n值，甚至相关侵蚀律本身仍存在争议。
参数变材料物质（特别是土壤质地和密度性），而实际力和会大量影响很大，空异可达2-3个数量级。
因此，在缺乏对潜数据时应通过敏感性分析采用考虑不同值性。
Hanson等人(2010)推荐的范围和通常用于制定这些参数的初步范围。
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堤切通常是由于堤切表面的侵蚀，其扩展也归因于侧壁崩塌的增加。
当每个单元的局部剪切应力超过临界值时，该单元被视为堤切。
堤切表面的侵蚀速率通常与超过临界值的剪切应力成正比。
这种关系通常使用幂律关系来描述，其中侵蚀速率是剪切应力的一个函数，该函数的指数通常在1到3之间变化。
这种关系的准确形式和参数值对于预测堤切的发展速率至关重要。
堤切的发展过程通常分为几个阶段，包括初始侵蚀、堤口扩大和最终溃决。
每个阶段的主导机制和相关的挑战可能有所不同。
例如，在初始阶段，侵蚀可能主要受到表面粗糙度和局部流动不稳定性的影响。
随着堤口的扩大，侧壁的稳定性变得越来越重要，重力驱动的侧壁崩塌可能成为主要的侵蚀机制。
最终，堤口的快速扩大和溃决过程可能受到下游水位和流量的影响。
为了模拟这些复杂的过程，通常需要采用能够考虑这些不同阶段和机制的综合模型。
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土壤的物理和力学特性，如粒径分布、密度、粘聚力和内摩擦角，对堤切的稳定性和侵蚀速率有显著影响。
水流的特性，如流速、剪切应力和湍流强度，直接驱动着堤切的侵蚀过程。
堤切的几何形状，如堤口的宽度、深度和侧壁的坡度，也会影响水流的模式和侵蚀的速率。
堤切的发展过程通常可以分为几个阶段，每个阶段都有其独特的特征和主导机制。
初始阶段可能涉及表面侵蚀和小规模的侧壁崩塌，而后期阶段则可能涉及大规模的侧壁崩塌和堤口的快速扩大。
理解这些阶段之间的转换对于准确预测堤切的发展至关重要

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堤切的初始形状和尺寸对后续的发展过程有着重要的影响。
例如，较大的初始堤口尺寸通常会导致更快的溃决速率和更大的峰值流量。
堤体的材料特性，包括其抗侵蚀能力和孔隙水压力，也起着关键作用。
此外，堤切周围的地形和水文条件也会显著影响其发展过程。
例如，上游水库的容量和下游河道的过流能力可以决定堤口扩大的速率和最终的峰值流量。
因此，为了准确预测堤切的发展过程，需要综合考虑这些复杂的因素及其相互作用。
堤切模型的发展历程
堤切模型的发展经历了从简单的经验公式到复杂的物理过程模型的演变。
早期的堤切模型通常基于简单的经验关系，例如堤口宽度随时间的变化或流量与水位之间的关系。
这些模型通常易于使用，但其适用范围有限，并且可能无法准确地反映堤切的复杂物理过程。
随着对堤切机理的深入了解和计算能力的提高，研究人员开发了更为复杂的物理过程模型。
这些模型通常基于质量守恒和动量守恒方程，并结合了堤体侵蚀、泥沙输移和堤口扩展等过程的描述。
一些著名的堤切模型包括DAMBRK模型、BREACH模型和WinDAM模型。
这些模型在不同的研究和应用中得到了广泛的应用和验证。
然而，这些模型也存在一些局限性，例如对某些参数的敏感性较高，以及对复杂地形和水流条件的适应性有限。
堤切模型的应用
堤切模型在洪水风险管理、水库调度和堤防设计等领域具有重要的应用价值。
例如，堤切模型可以用于预测堤切发生后的洪水淹没范围和水深，为制定应急预案提供依据。
堤切模型还可以用于评估不同堤防设计方案的安全性，为优化堤防设计提供参考。
此外，堤切模型还可以用于研究堤切对河流生态环境的影响，例如对鱼类栖息地和水质的影响。
堤切模型的发展趋势
堤切模型的发展趋势主要体现在以下几个方面：
一是提高模型的精度和可靠性，例如通过改进堤体侵蚀模型和泥沙输移模型；
二是扩展模型的适用范围，例如开发能够模拟复杂地形和水流条件的堤切模型；
三是提高模型的计算效率，例如采用并行计算技术和简化模型；
四是加强模型与实际应用的结合，例如开发基于模型的决策支持系统。
堤切模型的局限性
堤切模型的局限性主要体现在以下几个方面：
一是模型参数的确定困难，例如堤体侵蚀参数和泥沙输移参数；
二是模型对复杂地形和水流条件的适应性有限，例如对于具有复杂地形和水流条件的堤切，模型的模拟结果可能与实际情况存在较大偏差；
三是模型的计算效率较低，例如对于大规模的堤切模拟，模型的计算时间可能较长。
堤切模型的未来发展方向
堤切模型的未来发展方向主要包括以下几个方面：
一是开发基于物理过程的堤切模型，例如考虑堤体侵蚀、泥沙输移和堤口扩展等过程的耦合；
二是开发能够模拟复杂地形和水流条件的堤切模型，例如采用非结构网格和并行计算技术；
三是开发基于人工智能的堤切模型，例如利用机器学习和深度学习技术来预测堤切的发展过程。
堤切模型的应用案例
堤切模型的应用案例主要包括以下几个方面：
一是对历史堤切事件的模拟和分析，例如对1998年长江堤切事件的模拟和分析；
二是对未来堤切风险的评估和预测，例如对某个水库大坝的堤切风险进行评估和预测；
三是对堤防设计方案的评估和优化，例如对某个堤防工程的设计方案进行评估和优化

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Water Resources Research

RESEARCH ARTICLE
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Integration of a Levee Breach Erosion Model in a GPU-Accelerated 2D Shallow Water Equations Code

S. Dazzi[1], R. Vacondio[1], and P. Mignosa[1]

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Key Points:
- The paper presents a 2D SWE model able to simulate the opening of a levee breach due to overtopping and the subsequent flooding
- The physically based erosion model allows predicting the breach evolution in levees made of cohesive or noncohesive material
- The model can be applied to real cases, thanks to the fast execution times guaranteed by the GPU parallelization of computations

Supporting Information:
- Supporting Information S1

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Abstract This paper presents a two-dimensional shallow water equations code coupled with a physically based erosion model, able to predict the opening and evolution of breaches forming in levees built with either cohesive or noncohesive material. The bottom elevation change is evaluated using an excess shear-stress equation, which accounts for the hydrodynamic conditions and for the material characteristics. The proposed model modifies the local topography at runtime wherever the levee is overtopped without having to predefine the position and shape of the breach. The model is implemented in CUDA programming language, so that simulations can be run on graphics processing units, guaranteeing fast execution times even for high-resolution meshes and large domains. The validation is performed based on several experimental tests, and numerical predictions are in good agreement with the measurements. The strengths and weaknesses of the proposed approach are also discussed by comparison with a sediment transport model based on the Exner equation: While the latter gives good results only for breaches forming in levees built with noncohesive material, the proposed model can also be applied to cohesive embankments. The application to a historical flood event is also presented, showing that the model can effectively be employed for real field simulations also in the case of multiple breaches.

1. Introduction

Many lowland areas are protected from flooding by levees, whose main purpose is to confine the flow inside the river region. Levees are usually built with erodible material (clay, silt, or sand), and their height is designed to contain a specific hydrograph. However, in the case of an extreme event exceeding their design return period, overtopping occurs, often causing the embankment collapse. Land use modifications and climate change might also increase the discharge for a given return period, making the levees no longer adequate. Moreover, earthen embankments may experience breaching for piping and internal erosion processes, even before the water surface elevation reaches the levee crown. The dens of burrowing animals (e.g., porcupine, badger, and nutria) have been recently identified as another cause for levee collapse in Northern Italy (Orlandini et al., 2015; Sofia et al., 2017; Viero et al., 2013).

Since embankment failures can have damaging consequences and lead to huge economic and human losses, the design of flood hazard maps and emergency plans is particularly important, and numerical modeling represents a powerful tool for these analyses. The simplest approach to model the flood propagation caused by a levee breach is to adopt a one-dimensional (1D) model for the channel and a two-dimensional (2D) model for the floodable area outside the river region (e.g., Masoero et al., 2013; Vorogushyn et al., 2010). The breaches are included in the 1D model as lateral spillway structures (Mazzoleni et al., 2014), and the outflow from these structures is imposed as upstream boundary condition to the 2D model, often neglecting possible backwater effects caused by the presence of road or railway embankments in the floodable region. Moreover, the 1D model cannot accurately describe the markedly 2D flow near the breach inside the river. On the contrary, when both the river and the floodable area are simulated using fully 2D shallow water equations (SWE) models (e.g., Teng et al., 2017), these features can inherently be taken into account. In the past, the choice of using 1D-2D models was often necessary to reduce the computational effort, but the same outcome can now be achieved by using modern high-performance computing clusters and/or adopting parallelization techniques, such as implementing codes able to run on graphics processing units (GPUs; Dazzi et al., 2018; Vacondio et al., 2014, 2017).

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Water Resources Research

研究论文
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在GPU加速的二维浅水方程模型中集成堤防溃口侵蚀模型
方程模型

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关键要点：
- 本文提出一种二维浅水方程模型，能够模拟因漫顶导致的堤防溃口开启及其后续洪水演进
- 基于物理机制的侵蚀模型可预测由黏性或非黏性材料构筑的堤防溃口演化过程
- 得益于GPU计算并行化带来的快速执行能力，该模型可应用于实际案例

支持信息：
- 支持信息S1

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摘要：本文提出一种二维浅水方程模型，耦合基于物理机制的侵蚀模型，可预测由黏性或非黏性材料构筑的堤防所形成溃口的开启与演化过程。河床高程变化采用超量剪应力方程进行评估，该方程综合考虑水动力条件与材料特性。所提出的模型在运行过程中动态修正局部地形，一旦堤防发生漫顶即自动调整，无需预先设定溃口位置与形状。该模型采用CUDA编程语言实现，因而可在图形处理器（GPU）上运行，即使针对高分辨率网格与大范围区域亦能保障快速计算。模型验证基于多项试验，数值预测结果与实测数据吻合良好。本文还通过与基于Exner方程的泥沙输运模型对比，讨论了所提方法的优势与局限：后者仅对由非黏性材料构筑的堤防溃口具有良好模拟效果，而所提模型还可适用于黏性堤防。文中亦展示了该模型对一次历史洪水事件的应用，表明其亦可有效用于真实场景的模拟，包括多溃口情形。

1. 引言

许多低地地区依靠堤防抵御洪水，其主要功能是将水流约束于河道范围内。堤防通常由可冲刷材料（如黏土、粉土或砂）筑成，其高度按特定水文过程线设计。然而，在遭遇超过其设计重现期的极端事件时，将发生漫顶，往往导致堤防坍塌。土地利用变化与气候变化也可能提高特定重现期下的洪峰流量，致使现有堤防不再适用。此外，土质堤防甚至在水面高程尚未达到堤顶前，就可能因管涌与内部侵蚀过程而发生溃口。掘穴动物（如豪猪、獾和海狸鼠）的洞穴，近期已被确认为意大利北部堤防坍塌的另一诱因（Orlandini等，2015；Sofia等，2017；Viero等，2013）。

鉴于堤防失事可能造成严重后果，并导致巨大的经济损失与人员伤亡，洪水危险图绘制与应急预案制定尤为重要，而数值模拟则是开展此类分析的有力工具。模拟堤防溃口所致洪水演进的最简方法，是采用一维（1D）模型模拟河道，同时采用二维（2D）模型模拟河道以外的可淹没区域（例如Masoero等，2013；Vorogushyn等，2010）。溃口在1D模型中被作为侧向溢流结构处理（Mazzoleni等，2014），其出流作为上游边界条件施加于2D模型，但通常忽略可淹没区域内道路或铁路路基可能引起的回水效应。此外，1D模型无法准确描述溃口附近河道内显著的二维流动。相反，若采用全二维浅水方程（SWE）模型（例如Teng等，2017）同时模拟河道与可淹没区域，则上述特征可自然纳入考虑。过去，选择1D-2D模型常出于降低计算量的需要，但如今借助现代高性能计算集群和/或并行化技术（例如在图形处理器GPU上运行的代码实现；Dazzi等，2018；Vacondio等，2014，2017），同样可达成相同效果。

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The breach opening must be somehow included in the 2D modeling. The detailed simulation of the breach process is actually rather difficult, due to its three-dimensional nature and to the complex interactions between hydrodynamic conditions, bank stability, sediment transport, and infiltration processes; however, for practical use in the field of flood simulation, simplified approaches are often introduced. The gradual breach opening can be modeled by a time-varying topography with predefined geometric characteristics (e.g., Dewals et al., 2011) or by coupling the SWE with a 2D sediment transport model (e.g., Faeh, 2007).

For example, Vacondio et al. (2016) recently simulated a breach-generated flood with a GPU-accelerated fully 2D model, adopting a purely geometric approach to describe the breach evolution (specifying the breach position and assuming a trapezoidal shape, with final width and failure time defined a priori). In this way, however, the levee material characteristics are completely neglected from the breach modeling, possibly resulting in inaccurate predictions. The hydrodynamic conditions, such as the upstream inflow in the river, the presence of river bends and floodplains, possible backwater effects that reduce the velocity of outflowing water (and consequently the erosion), can also severely affect the breach evolution (Viero et al., 2013). Moreover, for (lateral) fluvial breaches the choice of "geometric" parameters cannot be assisted by results provided by the parametric (e.g., Froehlich, 2008; Xu & Zhang, 2009) or simplified (e.g., Chen & Anderson, 1987; Fread, 1988; Macchione, 2008; Mohamed et al., 2002; Visser, 1999; Wu, 2013) models available in the literature (ASCE/EWRI Task Committee on Dam/Levee Breaching, 2011), which were developed for (frontal) dam breach configurations. In fact, recent experimental investigations (Elalfy et al., 2017; Kakinuma & Shimizu, 2014; Michelazzo et al., 2018; Rifai et al., 2017; Wei et al., 2016) show that, apart from the very first stages, the evolution of a lateral breach is quite different from what observed in the frontal configuration, due to the different direction of the flow (parallel/tilted versus perpendicular to the levee crest).

The best option for simulating riverine levee breaches would be the use of detailed physically based multidimensional models, based on the integration of hydrodynamic and morphodynamic equations (e.g., Canelas et al., 2013; Li & Duffy, 2011; Murillo & García-Navarro, 2010). In this way, the flow along the river, through the breach, and in the inundated region can be simulated simultaneously without the need of introducing internal boundary conditions at the breach location. Attempts of applying 2D sediment transport models to simulate breaches in dams built with noncohesive material include those of Cao et al. (2011), Evangelista (2015), Guan et al. (2014), Juez et al. (2014), Van Emelen et al. (2015), Volz et al. (2012), Wang and Bowles (2006), and Wu et al. (2012), while Faeh (2007) performed a fluvial breach test. Nevertheless, applications to real cases are limited by the heterogeneity and scarcity of data on levee materials and by the prohibitive computational time required for running these models (unless parallelization techniques are introduced, see Juez et al., 2016). Moreover, the application of these models to breach erosion may be questionable, since most sediment transport equations were derived in uniform flow conditions, for small slopes, and with noncohesive materials, while the breach development is highly unsteady and often involves high slopes and cohesive sediments. For this reason, Morris et al. (2009) suggested that the employment of erosion laws would be more consistent with the breach process. Erosion laws have been applied to dam breach modeling by Chen and Anderson (1987), Morris et al. (2009), and Wang and Bowles (2006) and have the advantage of including specific erodibility parameters in the computations and of being applicable also to cohesive embankments. In fact, the erosion process of dams/levees built with noncohesive and cohesive material is quite different (e.g., Morris et al., 2007). In the latter case, headcut erosion is observed: One or more rills develop into a series of overfalls, which form a headcut (i.e., a vertical or nearly vertical drop on the bed); the headcut migrates upstream and reduces the dam crest height; this phase is then followed by a breach widening stage. Clearly, this three-dimensional process cannot be adequately simulated using a 2D depth-averaged model and sediment transport equations; hence, a headcut migration rate is often introduced to model this type of breaches (e.g., Hanson et al., 2005; Wu, 2013). However, since this quantity depends on the same erodibility coefficients that appear in simple erosion laws, such laws can arguably be used to model the general breach process, at least as regards the failure time and the final width, avoiding a detailed description of the headcut migration process.

The present work aims at introducing an efficient numerical tool for the simulation of inundations generated by levee breaches, including a physically based prediction of the breach evolution (instead of a geometric approach) and avoiding the necessity of defining its characteristics a priori. The GPU-accelerated 2D SWE numerical code PARFLOOD (Vacondio et al., 2014, 2017) was coupled with an erosion model, in which an excess shear-stress law is employed to predict the time evolution of the bottom elevation at the

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溃口的开启必须以某种方式纳入二维建模中。由于溃口过程具有三维特性，且水动力条件、岸坡稳定性、泥沙输运及渗流过程之间存在复杂相互作用，因此对其开展详细模拟实际上相当困难；然而，在洪水模拟的实际应用中，常引入简化方法。渐进式溃口开启可通过具有预设几何特征的时间变化地形（例如Dewals等，2011）或通过将浅水方程与二维泥沙输运模型耦合（例如Faeh，2007）来模拟。例如，Vacondio等（2016）最近采用GPU加速的全二维模型模拟了一次溃口引发的洪水，但采用纯几何方法描述溃口演化（指定溃口位置并假定为梯形形状，最终宽度和溃决时间均预先设定）。然而，这种方法完全忽略了堤防材料特性对溃口演化的贡献，可能导致预测结果不准确。上游河道入流、河道弯道与洪泛区的存在、可能存在的回水效应（降低出流流速，从而削弱侵蚀作用）等水动力条件，亦会显著影响溃口演化过程（Viero等，2013）。此外，对于（侧向）河流溃口，其几何参数的选择无法借助文献中已有的参数化模型（例如Froehlich，2008；Xu & Zhang，2009）或简化模型（例如Chen & Anderson，1987；Fread，1988；Macchione，2008；Mohamed等，2002；Visser，1999；Wu，2013）（美国土木工程师学会/环境与水资源研究所大坝/堤防溃口任务组，2011）所提供的成果进行辅助，因为这些模型是为（正向）大坝溃口工况开发的。事实上，近期实验研究表明（Elalfy等，2017；Kakinuma & Shimizu，2014；Michelazzo等，2018；Rifai等，2017；Wei等，2016），除初始阶段外，侧向溃口的演化过程与正向溃口观测结果明显不同，原因在于水流方向不同（平行或倾斜于堤顶，而非垂直于堤顶）。模拟河岸堤防溃口的最佳方案，应是采用基于水动力学与地貌动力学方程耦合的详细物理机制多维模型（例如Canelas等，2013；Li & Duffy，2011；Murillo & García-Navarro，2010）。如此一来，可同步模拟河道内水流、溃口处水流及淹没区内水流，无需在溃口位置引入内部边界条件。已有若干研究尝试将二维泥沙输运模型应用于非黏性材料构筑的大坝溃口模拟，包括Cao等（2011）、Evangelista（2015）、Guan等（2014）、Juez等（2014）、Van Emelen等（2015）、Volz等（2012）、Wang与Bowles（2006）以及Wu等（2012）的工作，而Faeh（2007）则开展了河流溃口试验。然而，这些模型在实际案例中的应用受限于堤防材料数据的异质性与稀缺性，以及运行此类模型所需的极高计算耗时（除非引入并行化技术，参见Juez等，2016）。此外，将这些模型应用于溃口侵蚀过程亦存疑，因为大多数泥沙输运公式是在均匀流、小坡度及非黏性材料条件下推导得出，而溃口发展过程高度非恒定，且常涉及陡坡与黏性泥沙。正因如此，Morris等（2009）建议采用侵蚀定律更契合溃口过程。Chen与Anderson（1987）、Morris等（2009）以及Wang与Bowles（2006）已将侵蚀定律应用于大坝溃口建模，其优势在于计算中可纳入特定的可蚀性参数，并适用于黏性堤坝。事实上，由非黏性与黏性材料构筑的大坝/堤防的侵蚀过程存在显著差异（例如Morris等，2007）。在后者情况下，可观察到溯源侵蚀现象：一条或多条细沟发育成一系列跌水，进而形成溯源侵蚀陡坎（即河床处的垂直或近似垂直落差）；该陡坎向上游迁移并降低坝顶高程；此后进入溃口展宽阶段。显然，这一三维过程无法通过二维垂向平均模型及泥沙输运方程得到充分模拟；因此，常引入溯源侵蚀陡坎迁移速率来模拟此类溃口（例如Hanson等，2005；Wu，2013）。然而，由于该迁移速率依赖于与简单侵蚀定律中相同的可蚀性系数，故此类定律原则上可用于模拟整体溃口过程，至少在溃决时间与最终宽度方面，而无需对溯源侵蚀陡坎迁移过程进行详尽描述。

本研究旨在引入一种高效的数值工具，用于模拟由堤防溃口引发的洪水，该工具采用基于物理机制的方法预测溃口演化过程（而非几何方法），并避免了预先定义溃口特征的必要性。本研究将GPU加速的二维浅水方程（SWE）数值模型PARFLOOD（Vacondio等，2014，2017）与一个侵蚀模型耦合，其中采用超量剪应力定律预测溃口处河床高程随时间的变化

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levee breach site as a function of the local hydrodynamic conditions and of the material characteristics. The model can be used for either noncohesive or cohesive embankment. For comparison purposes, a simple but robust bedload transport model (Juez et al., 2014) was implemented as well. Both models are enriched by a bank failure algorithm, which simulates the sudden failure of blocks of material due to slope instability. Validation is performed based on four experimental tests, and an example of application to the real levee-breach event that occurred on the Enza River (Italy) in December 2017 is also presented.

The paper is structured as follows. Section 2 presents the main features of the proposed model. Section 3 is dedicated to the description of all the test cases used for the validation and application of the model, while in section 4 a discussion on the advantages and disadvantages of the model is presented. In the last section, conclusions are drawn.

2. Model Description

本文结构如下:第二节介绍所提出模型的主要特征;第三节详细描述用于模型验证与应用的所有测试案例;第四节对模型的优缺点进行讨论;最后一节为结论部分

水动力模块+新开发的侵蚀和岸坡破坏模拟模块
还实现了一个泥沙输运模型

The proposed model couples three modules: a hydrodynamic model (already presented in previous work; see Vacondio et al., 2014, 2017) and two newly developed models for erosion and bank failure simulations. Moreover, a sediment transport model was implemented for comparison purposes (see Appendix A).

2.1. Hydrodynamic Model

The hydrodynamic model (Vacondio et al., 2014, 2017) is based on the 2D SWEs written in integral form (Toro, 2001) as follows:

$$\frac{\partial}{\partial t}\int_A \mathbf{U}\,\mathrm{d}A + \int_C \mathbf{H}\cdot\mathbf{n}\,\mathrm{d}C = \int_A \left(\mathbf{S}_0 + \mathbf{S}_f\right)\,\mathrm{d}A, \tag{1}$$

where A and C are the integration volume area and boundary, respectively, $\mathbf{U}$ is the vector of conserved variables, $\mathbf{H} = (\mathbf{F}, \mathbf{G})$ is the tensor of fluxes in the x and y directions, $\mathbf{n}$ is the outward unit vector normal to C, $\mathbf{S}_0$ and $\mathbf{S}_f$ are the bed and friction slope source terms, respectively, and t is the time. The well-balanced form of the SWEs, proposed by Liang and Marche (2009), is adopted:

$$\mathbf{U} = \begin{bmatrix} \eta \\ uh \\ vh \end{bmatrix}, \quad \mathbf{F} = \begin{bmatrix} uh \\ u^2h + \frac{1}{2}g\left(\eta^2 - 2\eta z\right) \\ uvh \end{bmatrix}, \quad \mathbf{G} = \begin{bmatrix} vh \\ uvh \\ v^2h + \frac{1}{2}g\left(\eta^2 - 2\eta z\right) \end{bmatrix}, \tag{2a}$$

$$\mathbf{S}_0 = \begin{bmatrix} 0 \\ -g\eta\frac{\partial z}{\partial x} \\ -g\eta\frac{\partial z}{\partial y} \end{bmatrix}, \quad \mathbf{S}_f = \begin{bmatrix} 0 \\ -gh\frac{n_f^2 u\sqrt{u^2+v^2}}{h^{4/3}} \\ -gh\frac{n_f^2 v\sqrt{u^2+v^2}}{h^{4/3}} \end{bmatrix}, \tag{2b}$$

In equation 2, $\eta = h + z$ is the water surface elevation, h is the flow depth, and z is the bed elevation; u and v are the velocity components along the x and y directions, respectively, n_f is Manning's roughness coefficient, and g is the acceleration due to gravity.

An explicit finite-volume scheme is used to discretize the equations; both first-order and second-order accurate approximations in space and time are implemented, but only the first-order approximation is here recalled for the sake of brevity. The following equation is used to update the conserved variables in time:

$$\mathbf{U}_{i,j}^{n+1} = \mathbf{U}_{i,j}^{n} - \frac{\Delta t}{\Delta x}\left(\mathbf{F}_{i+\frac{1}{2},j} - \mathbf{F}_{i-\frac{1}{2},j}\right) - \frac{\Delta t}{\Delta y}\left(\mathbf{G}_{i,j+\frac{1}{2}} - \mathbf{G}_{i,j-\frac{1}{2}}\right) + \Delta t\left(\mathbf{S}_0 + \mathbf{S}_f\right). \tag{3}$$

Subscripts i, j represent the cell position, while superscript n refers to the time level; Δx and Δy are the cell dimensions in the x and y directions, respectively, and Δt is the time step size. The Harten-Lax-van Leer contact (HLLC) approximate Riemann solver (Toro, 2001) is used to compute intercell fluxes, and the correction proposed by Kurganov and Petrova (2007) is implemented to avoid nonphysical velocity values at wet/dry fronts. The slope source term is discretized with a centered approximation (Vacondio et al., 2014), while the friction source term is discretized using the implicit formulation proposed by

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溃口位置取决于当地水动力条件及材料特性。本模型既适用于无黏性堤防，也适用于黏性堤防。为便于对比，亦实现了一种简单但稳健的推移质输运模型（Juez 等，2014）。两个模型均集成了岸坡失稳算法，可模拟因边坡失稳导致的材料块体突发性坍塌。模型验证基于四项试验，同时还给出了该模型应用于 2017 年 12 月意大利恩扎河真实溃堤事件的实例。

本文结构如下：第 2 节介绍所提模型的主要特征；第 3 节详述用于模型验证与应用的所有测试算例；第 4 节讨论该模型的优缺点；最后一节给出结论。

2. 模型描述

本文结构如下:第二节介绍所提出模型的主要特征;第三节详细描述用于模型验证与应用的所有测试案例;第四节对模型的优缺点进行讨论;最后一节为结论部分

水动力模块+新开发的侵蚀和岸坡破坏模拟模块
还实现了一个泥沙输运模型

所提出的模型耦合了三个模块：一个水动力模型（已在前期工作中提出；参见 Vacondio 等，2014，2017）以及两个新开发的侵蚀与岸坡失稳模拟模型。此外，还实现了一个泥沙输运模型用于对比分析（见附录 A）。

2.1 水动力模型

水动力模型（Vacondio 等，2014，2017）基于以积分形式表达的二维浅水方程（SWEs）（Toro，2001），具体如下：

$$\frac{\partial}{\partial t}\int_a U\,dA + \int cH\cdot n\,dC = \int_a (S + Sf)\,dA; \tag{1}$$

其中 A 和 C 分别为积分区域面积与边界，U 为守恒变量向量，H = (F, G) 为 x 与 y 方向上的通量张量，n 为垂直于边界 C 的外法向单位矢量，S 与 Sf 分别为床面坡度与摩阻坡度源项，t 为时间。本文采用 Liang 与 Marche（2009）提出的浅水方程（SWEs）良好平衡形式：

$$\mathbf{U} = \begin{bmatrix} \eta \\ uh \\ vh \end{bmatrix}; \quad F = \begin{bmatrix} uh \\ u^2h + \tfrac{1}{2}g(\eta^2 - 2\eta z)uvh \end{bmatrix}; \quad G = \begin{bmatrix} vh \\ uvh \\ v^2h + \tfrac{1}{2}g(\eta^2 - 2\eta z) \end{bmatrix}, \tag{2a}$$

$$S = \begin{bmatrix} 0 \\ -g\eta\frac{\partial z}{\partial x} \\ -g\eta\frac{\partial z}{\partial y} \end{bmatrix}; \quad Sf = \begin{bmatrix} 0 \\ -gh\frac{n^2 f\, u\sqrt{u^2+v^2}}{h^{4/3}} \\ -gh\frac{n^2 f\, v\sqrt{u^2+v^2}}{h^{4/3}} \end{bmatrix}, \tag{2b}$$

在公式（2）中，η = h + z 为水面高程，h 为水流深度，z 为河床高程；u 和 v 分别为沿 x 和 y 方向的速度分量；nf 为曼宁糙率系数，g 为重力加速度。

采用显式有限体积法对控制方程进行离散；空间与时间上分别实现了精度为一阶和二阶的近似格式，但此处仅简要回顾一阶近似格式。以下方程用于更新守恒变量的时间演化：

$$\mathbf{U}_{i,j}^{n+1} = \mathbf{U}_{i,j}^{n} - \frac{\Delta t}{\Delta x}\left(\mathbf{F}_{i+\frac{1}{2},j} - \mathbf{F}_{i-\frac{1}{2},j}\right) - \frac{\Delta t}{\Delta y}\left(\mathbf{G}_{i,j+\frac{1}{2}} - \mathbf{G}_{i,j-\frac{1}{2}}\right) + \Delta t(\mathbf{S}_0 + \mathbf{S}_f). \tag{3}$$

下标 i、j 表示网格单元位置，上标 n 表示时间层；Δx 和 Δy 分别为 x 和 y 方向上的网格单元尺寸，Δt 为时间步长。采用 Harten-Lax-van Leer 接触（HLLC）近似黎曼求解器（Toro，2001）计算单元间通量，并引入 Kurganov 与 Petrova（2007）提出的修正方法，以避免湿/干交界面处出现非物理解速度值。坡度源项采用中心差分格式离散（Vacondio 等，2014），而摩阻源项则采用……提出的隐式格式离散

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Caleffi et al. (2003). The minimum allowable time step is computed according to the Courant-Friedrichs-Lewy stability condition (Toro, 2001). The model is implemented in a CUDA/C++ code, which exploits the intrinsic parallelization of computations on GPU devices, thus guaranteeing fast execution times compared to serial codes. More details on the scheme and implementation can be found in Vacondio et al. (2014, 2017).

2.2. Erosion Model 侵蚀模型

The breach evolution is predicted according to the erosion model described in the following. The bottom elevation change is not allowed in the whole domain but only along the erodible levees, where potential breaches might occur. In these selected regions, the morphodynamic evolution is described by the following equation:

$$\frac{\partial z}{\partial t} = -\frac{E}{1-p}, \tag{4}$$

where p is the bed porosity and E is the bed erosion rate (eroded volume per unit area per unit time). This latter quantity can be estimated according to a linear erosion law, also referred to as excess stress equation (e.g., Hanson & Simon, 2001; Partheniades, 1971):

床面侵蚀速率

$$E = \begin{cases} k_d(\tau - \tau_c) & \text{if } \tau > \tau_c, \\ 0 & \text{if } \tau \le \tau_c. \end{cases} \tag{5}$$

In equation (5), k_d represents the erodibility coefficient of the embankment material, while τ and τ_c are the bed shear stress and its critical value for sediment motion, respectively. The bed shear stress is computed as follows:

床面剪应力

$$\tau = \gamma \frac{n_f^2(u^2 + v^2)}{h^{1/3}} \tag{6}$$

where γ is the specific weight of water. Equation (4) is simply discretized as follows:

$$z_{i,j}^{n+1} = z_{i,j}^{n} - \frac{E(\tau_{i,j})}{1-p}\Delta t. \tag{7}$$

In essence, equation (7) modifies the bottom elevation of each cell (belonging to an erodible levee) at runtime according to the local value of the bed shear stress, and nonnegligible erosion is only obtained in the practice whenever and wherever the levee is overtopped. The opening of a breach, and its subsequent deepening and widening, is then predicted automatically, without the need to predefine the breach position and dimensions as input data. A minimum bed elevation can also be specified in order to avoid erosion below a nonerodible foundation. From equation (7), it can be noticed that only scour is allowed, while deposition of sediments is not included in the computations: The eroded material is supposed to be washed away by the flow, in line with other simplified approaches. Conversely, the global water volume is conserved by maintaining the original water depth of each cell where sediment is eroded; hence, the water surface elevation must be adjusted in these cells by subtracting the scour computed during the current update.

By means of equations (4) and (5), the breach erosion process depends both on the flow field characteristics (via τ) and on the embankment material (via k_d, τ_c, and p). The erodibility parameters k_d and τ_c control the erosion process and thus must be accurately calibrated. Despite the fact that specific experimental tests for their determination were developed, such as the jet erosion test (Hanson & Cook, 2004), reliable correlations with the sediment characteristics are not available yet. Different test configurations and interpretations of results lead to different estimates for k_d and τ_c (Khanal et al., 2016), and even the use of a linear erosion law is still debated (Walder, 2015). Moreover, these parameters are observed to be quite sensitive not only to the type of material, in particular to soil texture and plasticity, but also to the compaction effort and water content (Fell et al., 2013; Nguyen et al., 2017; Wahl et al., 2009), with a variability up to 2–3 orders of magnitude (Hanson & Hunt, 2007). Therefore, the choice of erodibility parameters must be carefully considered,

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Caleffi 等（2003）。最小允许时间步长根据 Courant-Friedrichs-Lewy 稳定性条件（Toro，2001）计算得出。该模型采用 CUDA/C++ 编程实现，充分利用 GPU 设备计算的内在并行性，从而确保相较于串行代码更短的执行时间。有关该数值格式及其实现的更多细节，参见 Vacondio 等（2014，2017）。

2.2. 侵蚀模型　　侵蚀模型

溃口演化过程依据下述侵蚀模型进行预测。河床高程变化并非在整个计算域内允许，而仅限于可能发生溃口的可蚀堤防沿线区域。在这些选定区域内，地貌动力学演化由下列方程描述：

$$\frac{\partial z}{\partial t} = -\frac{E}{1-p}, \tag{4}$$

其中p为床面孔隙率，E为床面侵蚀速率（单位面积单位时间内的侵蚀体积）。该量可依据线性侵蚀定律（又称超额应力方程）估算（例如Hanson与Simon，2001；Partheniades，1971）：

床面侵蚀速率

$$E = \begin{cases} k_d(\tau - \tau_c), & \text{当} \tau > \tau_c; \\ 0, & \text{当} \tau \le \tau_c。 \end{cases} \tag{5}$$

(86)

在公式（5）中，kd表示堤防材料的可蚀性系数，τ与τc分别表示床面剪应力及其引发泥沙运动的临界值。床面剪应力按如下公式计算：

床面剪应力

$$\tau = \gamma \frac{n^2 f (u^2 + v^2)}{h^{1/3}} \tag{6}$$

其中γ为水的容重。公式（4）直接离散化如下：

$$z_{n+1}^{i,j} = z_n^{i,j} - \frac{E(\tau_{i,j})}{1-p}\Delta t。 \tag{7}$$

本质上，公式（7）根据各网格单元（属可蚀堤防）局部床面剪应力实时修正其河床高程，实践中仅当且仅当堤防发生漫顶时才产生不可忽略的侵蚀。溃口的开启及其后续加深与拓宽过程由此自动预测，无需预先设定溃口位置与尺寸作为输入数据。亦可指定最小河床高程，以避免侵蚀深入至不可蚀基础层。由公式（7）可见，计算仅允许冲刷，不包含泥沙淤积：被侵蚀物质假定随水流全部带走，这与其他简化方法一致。反之，通过维持各发生侵蚀网格单元原有的水深，整体水量得以守恒；因此，这些单元的水面高程需在本次更新所计算的冲刷量基础上相应下调。

依据公式（4）与（5），溃口侵蚀过程既取决于流场特征（通过τ体现），也取决于堤防材料特性（通过kd、τc与p体现）。可蚀性参数kd与τc主导侵蚀过程，故须精确率定。尽管已有专门试验方法用于测定这两项参数（如喷射侵蚀试验，Hanson与Cook，2004），但目前尚无可靠的相关性关系可将其与泥沙特性关联起来。不同试验配置与结果解读方式导致kd与τc的估算值存在差异（Khanal等，2016），甚至线性侵蚀定律本身仍存争议（Walder，2015）。此外，这些参数不仅对材料类型（尤其是土壤质地与塑性）高度敏感，亦显著受压实功与含水率影响（Fell等，2013；Nguyen等，2017；Wahl等，2009），其变化幅度可达2–3个数量级（Hanson与Hunt，2007）。因此，在选取可蚀性参数时必须审慎考量。

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and the uncertainty in their values must be properly taken into account by means of a sensitivity analysis when data for calibration are not available. The diagram and tables reported by Hanson et al. (2010) can provide guidance for defining the range of variability of these parameters. Moreover, some authors suggest empirical relationships between τ_c and k_d (e.g., Hanson & Simon, 2001; Nguyen et al., 2017); Wu (2013) also reports an empirical formula for computing the erodibility coefficient (based on the clay content and the dry specific weight of the soil), which can help in the choice of reasonable values when specific erosion tests cannot be performed.

2.3. Bank Failure Algorithm 堤岸失稳算法

While the breach triggering is mainly due to the erosion following levee overtopping, its enlargement is also due to the sudden collapse of the lateral banks, which lose stability as long as the breach deepens (Hunt et al., 2005). Numerical models must consider this mechanism for a correct prediction of the breach development. For this reason, different bank failure algorithms were presented in the literature (e.g., Evangelista et al., 2015; Swartenbroekx et al., 2010; Volz et al., 2012), usually based on the idea of reducing the local slope of each cell when it exceeds a critical value φ_c. Obviously, all these models ignore cantilever failures, which are sometimes observed in experimental tests (e.g., Wei et al., 2016) but cannot be described in a depth-averaged model. In this work, the scheme of Guan et al. (2014) for structured grids was adapted for guaranteeing efficiency on GPUs.

Two different values are specified for the critical angles, depending on the fact that the cell is above or below the water surface level: $\varphi_{c,\ \mathrm{wet}}$ for wet cells and $\varphi_{c,\ \mathrm{dry}}$ for dry cells; moreover, φ_{dep} is the angle that the deposited material forms after collapse. These values are often assumed equal to the angle of repose of the material. Let us consider cell (i, j) with bottom elevation $z_{i,\ j}$ in a Cartesian grid, with four neighbor cells $(i + 1, j)$, $(i - 1, j)$, $(i, j + 1)$, $(i, j - 1)$. The local slope φ_k in the kth direction can be computed as

$$\tan\varphi_k = \left(z_k - z_{i,\mathrm{j}}\right)/l_k, \tag{8}$$

where z_k is the bottom elevation of the neighboring cell in the kth direction (i.e., cell $(i + 1, j)$ to the east, cell $(i, j + 1)$ to the north, etc.), and l_k is the grid size in the same direction (i.e., Δx to the east/west and Δy to the south/north). If $|\varphi_k| > \varphi_c$ (wet or dry, depending on the cell state), then the bottom is considered locally unstable, and a correction Δz_k can be calculated as

$$\Delta z_k = 0.5\, l_k \left(\tan|\varphi_k| - \tan\varphi_{\mathrm{dep}}\right)\ \mathrm{sign}(\varphi_k). \tag{9}$$

An equal and opposite correction will be computed in the neighboring cell, so that the total sediment mass is conserved in this procedure, and the bed slope is simply tilted from φ_k to φ_{dep}. Finally, the updated value for the bottom elevation becomes

$$z_{i,j}^{\mathrm{new}} = z_{i,j} + \sum_{k=1}^{4} \Delta z_k. \tag{10}$$

These operations are performed in a specific CUDA kernel, and threads (i.e., the basic work unit in CUDA, corresponding to one computational cell) are processed in parallel. Differently from that by Guan et al. (2014), who processed all cells sequentially, and applied the correction Δz_k both to the current cell and to its neighbor (with proper sign), in this implementation each thread computes its own corrections, even at the cost of repeating calculations twice in two different threads. As already discussed in Vacondio et al. (2014) as regards intercell fluxes, accepting this small computational overhead makes the code more efficient than storing an extra array for the values of Δz_k and accessing it later.

Obviously, local changes in the bed slope may in turn affect the stability of other neighboring cells. The algorithm previously described, then, must be repeated iteratively until no more corrections are necessary. In order to reduce the computational time, this recursive procedure is not performed at every time step Δt dictated by the Courant-Friedrichs-Lewy condition but at a larger pace Δt_{stab}; in particular, preliminary tests showed that checking and correcting the slope stability with $\Delta t_{\mathrm{stab}} = 500 - 1{,}000\Delta t$ provides the same results as with $\Delta t_{\mathrm{stab}} = \Delta t$.

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当缺乏率定数据时，其取值不确定性须通过敏感性分析予以恰当考虑。汉森等（2010）所列图表与表格可为界定这些参数的变幅范围提供参考。此外，部分学者提出了τc与kd之间的经验关系式（例如Hanson与Simon，2001；Nguyen等，2017）；吴（2013）亦给出一个估算可蚀性系数的经验公式（依据黏粒含量与土壤干容重），当无法开展专门侵蚀试验时，该公式有助于选取合理参数值。

2.3. 堤岸失稳算法 堤岸失稳算法

尽管溃口触发主要源于堤防漫顶后的侵蚀，但其扩展亦由侧向堤岸的突发性坍塌所致——随着溃口加深，堤岸逐渐丧失稳定性（Hunt等，2005）。数值模型必须纳入此机制，方能准确预测溃口发展过程。因此，文献中提出了多种堤岸失稳算法（例如Evangelista等，2015；Swartenbroekx等，2010；Volz等，2012），通常基于如下思路：当某单元局部坡度超过临界值φc时，即对其局部坡度予以削减。显然，所有此类模型均忽略悬臂式失稳现象——该类现象虽偶见于试验（例如Wei等，2016），却无法在垂向平均模型中予以描述。本文工作将关等（2014）针对结构化网格提出的方案加以改进，以确保其在GPU平台上的计算效率。

临界角度φc取值分两种情形：单元位于水面以上时取φc, dry，位于水面以下时取φc, wet；此外，φdep为塌落沉积物形成的坡角，该值通常取为材料的休止角。现考虑笛卡尔网格中单元(i, j)，其河床高程为zi,j，具有四个相邻单元：(i + 1, j)、(i − 1, j)、(i, j + 1)、(i, j − 1)。第k方向的局部坡度φk可计算为

$$\tan\varphi k = (zk - zi;j)/lk; \tag{8}$$

其中zk为第k方向相邻单元的河床高程（即东侧单元(i + 1, j)、北侧单元(i, j + 1)等），lk为该方向的网格尺寸（即东西向为Δx，南北向为Δy）。若|φk| > φc（湿态或干态取决于单元状态），则认为该处床面局部失稳，可计算修正量Δzk为

$$\Delta zk = 0.5\, lk\, (\tan|\varphi k| - \tan\varphi dep)\, \text{sign}\, \varphi k\, ()\ 。\ (9)$$

相邻单元将计算大小相等、方向相反的修正量，从而在此过程中守恒总泥沙质量，且床面坡度仅由φk倾斜至φdep。最终，更新后的河床高程为

$$z^{new}_{i;j} = zi;j + \sum 4k=1 \Delta zk。\ (10)$$

上述运算在特定CUDA核函数中执行，线程（即CUDA中基本计算单元，对应一个计算网格单元）并行处理。与关等（2014）采用所有网格单元顺序处理、并在当前单元及其相邻单元上同时施加修正量Δzk（符号适当）的做法不同，本实现中每个线程独立计算自身修正量，即使这意味着同一计算可能在两个不同线程中重复执行两次。如瓦康迪奥等（2014）在讨论单元间通量时已指出，接受这一微小计算开销，反而比额外开辟一个数组用于存储Δzk值并在后续访问更为高效。

显然，局部床面坡度的变化可能继而影响邻近其他网格单元的稳定性。因此，前述算法需迭代执行，直至不再需要任何修正为止。为减少计算耗时，该递归过程并非按库朗–弗里德里希斯–勒维（CFL）条件所限定的每个时间步长Δt执行，而是以更大间隔Δtstab执行；具体而言，初步测试表明，当以Δtstab = 500–1000Δt的频率进行坡度稳定性检查与修正时，所得结果与Δtstab = Δt时完全一致。

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3. Numerical Tests

五个测试案例，用于验证所提出的侵蚀模型

In this section, five test cases are presented for the validation of the erosion model. The first case is a 1D frontal dam breach experiment and was chosen for its simplicity in order to highlight the differences in the predictions of the erosion model compared to a sediment transport model. Moreover, two 2D frontal breach test cases, which differ in the type of material used for building the dam (cohesive versus noncohesive), were considered in order to assess the capability of predicting the breach enlargement and to study the influence of the model parameters on the simulation results. Then, since the erosion model is conceived for real field applications to fluvial breaches, a large-scale experimental test case concerning a lateral breach in a channel was privileged over available small-scale laboratory experiments as the fourth validation test, and it still involves levees built with noncohesive material. Finally, the applicability of the model to a real event is assessed by means of the simulation of the levee breach on the Enza River (Northern Italy) and the subsequent flooding that occurred in December 2017.

3.1. One-Dimensional Frontal Dam Breach

一维正面溃坝

The experimental test concerning dam erosion due to overtopping, reported by Tingsanchali and Chinnarasri (2001; Test C-2), was simulated in order to compare the results of the erosion and the sediment transport models. A 0.8-m-high dam, with crest width equal to 0.3 m and upstream and downstream slopes equal to 1V:3H and 1V:2.5H, respectively, was built in a 35-m-long, 1-m-wide rectangular flume. The dam was built with sand with the following characteristics: $d_{50} = 0.86$ mm, $d_{30} = 0.52$ mm, $d_{90} = 3.80$ mm, $d_m = 1.13$ mm, and $\rho_s = 2{,}650$ kg/m^3. A constant inflow (1.23 L/s) was supplied at the upstream end of the channel; a vertical plate was held at the dam crest until the reservoir was filled, and the water depth in the reservoir was 3 cm higher than the dam crest; then, the sudden removal of the plate allowed overflow to start.

The domain was discretized by means of a uniform grid with $\Delta x = \Delta y = 0.05$ m. The following parameters were common to both erosion and sediment transport models: bed porosity $p = 0.4$, critical angles $\varphi_{c,\text{wet}} = \varphi_{\text{dep}} = 30°$, and $\varphi_{c,\text{dry}} = 50°$. Manning's coefficient and the erosion model parameters, τ_c and k_d, were subjected to calibration, and a sensitivity analysis on their values was performed.

首先，利用泥沙输运模型对实验进行模拟，该模型无需其他额外的率定参数，从而确定曼宁系数。

First, Manning's coefficient was determined by simulating the experiment with the sediment transport model, which did not require any other calibration coefficient. Figures 1a and 1b compare the experimental dam profiles along the centerline of the breach at selected times ($t = 30$ s and $t = 60$ s) and the results from the numerical simulations performed assuming $n_f = 0.016$, 0.018, and 0.020 m$^{-1/3}$·s. An increase in this coefficient leads to a slightly more rapid erosion. However, there is no clear best fit value; in the first phases (Figure 1a), the smallest roughness value seems to mimic the dam erosion better, while it underestimates the erosion at the dam crest after some time (Figure 1b). The intermediate value (0.018 m$^{-1/3}$·s) is hence selected for describing the general process.

The erosion model was tested next. As a starting point for choosing the erodibility parameters, the work by Hanson et al. (2010) was considered: For soils with low clay content, the suggested values for k_d are in the range 50–800 cm^3·N^{-1}·s^{-1} (depending on compaction); the corresponding τ_c range is 10^{-3}–10^{-1} Pa.

Preliminary simulations were performed varying τ_c (10^{-3}, 10^{-2}, and 10^{-1} Pa) with different values of k_d, and results of these tests show that changes in the values of τ_c have negligible effects in the process, because the bed shear stress exceeds the critical value by 1–2 orders of magnitude ($\tau \gg 1$). Then, τ_c was assumed equal to 10^{-2} Pa, and the erodibility coefficient was varied: The best fit was obtained for $k_d = 500$ cm^3·N^{-1}·s^{-1}. In Figures 2a and 2b, the simulated and measured dam profiles along the centerline at selected times for different values of k_d (300, 400, 500, and 600 cm^3·N^{-1}·s^{-1}) are reported and stress how much the choice of this parameter can influence the breach evolution. The fact that the proposed model can only reproduce the erosion process can be clearly noticed from this comparison: While experimentally, the downstream slope flattens as long as the dam crest is eroded because sediments are deposited at the dam toe, numerically, the downhill slope simply retreats due to erosion; nevertheless, the calibrated value for k_d guarantees that the model predicts the eroded dam crest height correctly. The outflow discharge is reproduced reasonably, and the peak value is well predicted, as can be observed in Figure 2c, even if the model, probably due to the differences in the bathymetry toward the end of the simulation (caused by the exclusion of the deposition

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3. 数值试验

五个测试案例，用于验证所提出的侵蚀模型

本节共呈现五个验证侵蚀模型的试验案例。第一个案例为一维正向溃坝试验，因其结构简单而被选中，旨在凸显侵蚀模型与泥沙输运模型在预测结果上的差异。此外，还考虑了两个二维正向溃坝试验案例，二者建坝材料类型不同（黏性土与非黏性土），用以评估模型对溃口扩展过程的预测能力，并研究模型参数对模拟结果的影响。接着，由于本侵蚀模型专为实际河流溃口工程应用而设计，故在可选的小型室内试验之外，优先选取了一个涉及渠道侧向溃口的大尺度试验案例作为第四项验证试验，该试验仍采用非黏性土修筑堤防。最后，通过模拟2017年12月意大利北部恩扎河（Enza River）堤防溃决及后续洪水事件，进一步评估该模型在真实事件中的适用性。

3.1. 一维正向溃坝

一维正面溃坝

丁桑查利和钦纳拉西（2001；试验C-2）所报道的关于漫顶导致坝体侵蚀的试验被进行了数值模拟，以对比侵蚀模型与泥沙输运模型的计算结果。试验在一条长35 m、宽1 m的矩形水槽中构建了一座高0.8 m的坝体，坝顶宽度为0.3 m，上下游边坡分别为1V:3H和1V:2.5H。坝体由砂料筑成，其颗粒特性如下：d50 = 0.86 mm，d30 = 0.52 mm，d90 = 3.80 mm，dm = 1.13 mm，ρs = 2650 kg/m³。水流以恒定流量（1.23 L/s）从水槽上游端引入；在水库蓄水完成前，于坝顶处竖立一块垂直挡板，使水库水位比坝顶高出3 cm；随后突然移除该挡板，从而引发溢流。

计算域采用均匀网格离散，Δx = Δy = 0.05 m。以下参数在侵蚀模型与泥沙输运模型中通用：床面孔隙率 p = 0.4，临界休止角 φc,wet = φdep = 30°，φc,dry = 50°。曼宁系数及侵蚀模型参数 τc 和 kd 需经率定，并对其取值开展敏感性分析。

首先，利用泥沙输运模型对实验进行模拟，该模型无需其他额外的率定参数，从而确定曼宁系数。

首先，通过采用无需其他率定系数的泥沙输运模型模拟该试验，确定曼宁系数。图1a 和 1b 对比了在若干典型时刻（t = 30 s 和 t = 60 s）沿溃口中心线的实测坝体剖面与数值模拟结果，其中曼宁系数 nf 分别取 0.016、0.018 和 0.020 $m^{1/3}·s$。该系数增大时，侵蚀速率略有加快。然而，并无明确的最优拟合值：在初期阶段（图1a），最小糙率值似乎更贴近实测坝体侵蚀过程；但一段时间后（图1b），其对坝顶侵蚀程度的预测则偏低。因此，选取中间值（0.018 $m^{1/3}·s$）来表征整体过程。

随后对侵蚀模型进行了验证。在确定可蚀性参数初始取值时，参考了 Hanson 等人（2010）的研究成果：对于低黏土含量土壤，建议的 kd 取值范围为 50–800 $cm^3·N^{1}·s^{1}$（取决于压实程度），相应 τc 范围为 10^3–10^1 Pa。初步模拟中，τc 分别取 10^3、10^2 和 10^1 Pa，并结合不同 kd 值进行组合试验；

结果表明，τc 取值变化对整个过程影响甚微，因为床面切应力远超临界值（$\tau \gg 1$），相差达1–2个数量级。因此，τc 取为 10^2 Pa，再对可蚀性系数 kd 进行调整：最优拟合结果对应 kd = 500 $cm^3·N^{1}·s^{1}$。图2a 和 2b 展示了不同 kd 值（300、400、500 和 600 $cm^3·N^{1}·s^{1}$）下，沿溃口中心线在若干典型时刻的模拟与实测坝体剖面，清晰反映出该参数选择对溃口演化过程的显著影响。从该对比中可明显看出，所提模型仅能再现侵蚀过程：实验中，随着坝顶被侵蚀，下游坡面因坝趾处泥沙淤积而趋于平缓；而数值模拟中，下游坡面仅因侵蚀而后退；不过，经率定的 kd 值可确保模型准确预测被侵蚀坝顶的高程。出流量整体再现效果合理，峰值流量预测良好（见图2c），尽管模型可能因模拟末期地形（因忽略淤积过程所致）差异，导致流量过程线的退水段被高估。

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曼宁系数增大，会使侵蚀过程略微加快，因此选择中间值0.018

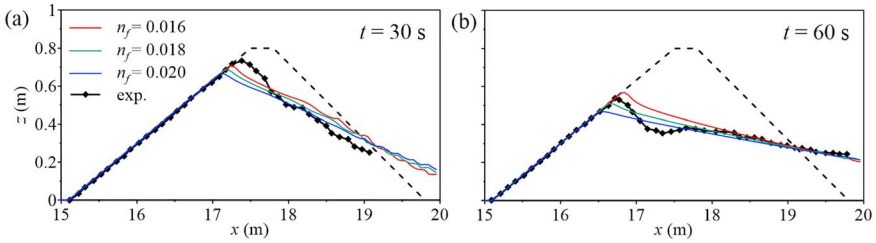


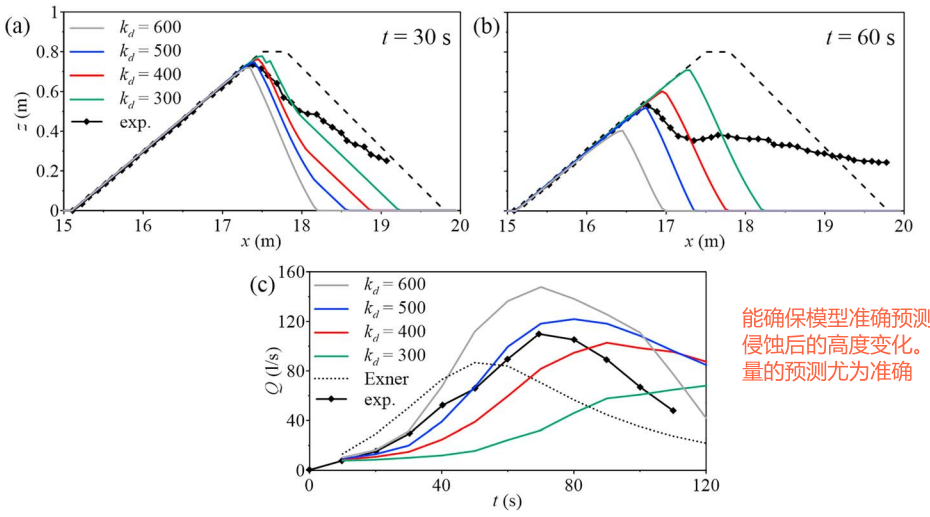
Figure 1. One-dimensional frontal dam breach test. Experimental dam profiles and numerical results with the sediment transport model, assuming different values for Manning's coefficient n_f (in $m^{-1/3}$·s), at (a) $t = 30$ s and (b) $t = 60$ s. The dashed line represents the initial dam profile.

processes), overestimates the falling limb of the hydrograph. The outflow discharge predicted by the sediment transport model is also reported in Figure 2c for the sake of comparison; the dam erosion is probably too fast in the first stages, and the reservoir emptying is anticipated compared to experimental observations, leading to a lower discharge peak.

3.2. Two-Dimensional Frontal Breaches 二维正向模型

In this section, a sensitivity analysis on the erosion model parameters is undertaken by simulating two 2D frontal breaches experimental test cases performed at the HR Wallingford laboratory for the IMPACT project (Morris & Hassan, 2005). All tests were carried out in a 50-m-long and 10-m-wide flume, where an erodible dam was built roughly 36 m downstream from the channel entrance (Figure 3a). Water was allowed into the flume until the reservoir upstream of the dam was filled and the water level exceeded the elevation of a pilot channel carved in the central portion of the dam, thus triggering the breach opening. Tests labeled #2 and #10 were selected, and the main features of these experiments are reported in Table 1. Notably, in Test #2 the dam was built with noncohesive material (nearly uniform sand with $d_{50} = 0.25$ mm), while for Test #10 a cohesive material (clay) was employed; the breach evolution was hence different for the two tests. For the noncohesive dam, the first phase was similar to that in the 1D case and was characterized by uniform erosion on the downhill slope, which retreated and became milder at the dam section corresponding to the pilot channel. On the other hand, for the test with a cohesive dam, headcut erosion was observed on the downhill slope. Then, in both cases, the breach side slopes started to lose stability and bank failures occurred, so that the breach enlarged in time symmetrically; the process ended when the upstream reservoir was almost empty.

500效果最佳



能确保模型准确预测坝顶被侵蚀后的高度变化。洪峰流量的预测尤为准确

Figure 2. One-dimensional frontal dam breach test. Sensitivity analysis on the erodibility coefficient k_d (in $cm^3 \cdot N^{-1} \cdot s^{-1}$) with the erosion model: Dam profiles at (a) $t = 30$ s and (b) $t = 60$ s and (c) overflow discharge. The discharge predicted by the Exner based model is also reported in (c) for comparison. The dashed line represents the initial dam profile.

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曼宁系数增大，会使侵蚀过程略微加快，因此选择中间值0.018

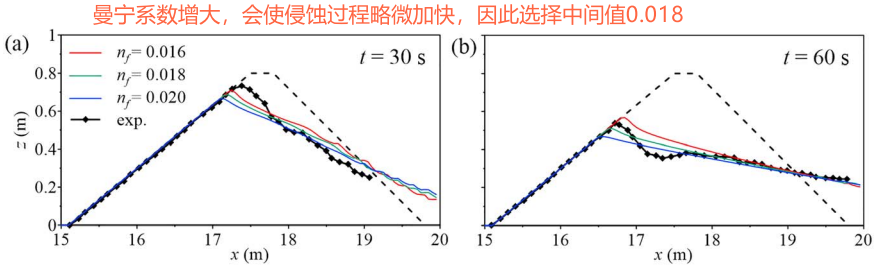


图1. 一维正向溃坝试验：实测坝体剖面与采用泥沙输运模型所得数值结果对比（假定不同曼宁系数 nf（单位：$m^{1/3}·s$）），分别对应（a）t = 30 s 和（b）t = 60 s 时刻。虚线表示初始坝体剖面。

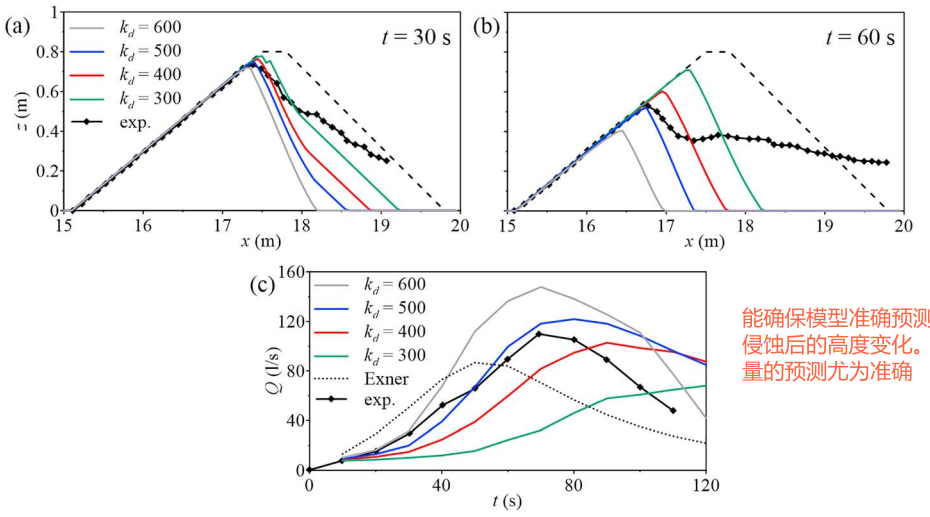
过程），导致流量过程线的退水段被高估。为便于对比，图2c 同时给出了泥沙输运模型预测的出流量；溃坝初期侵蚀速率可能过快，水库排空时间较实测提前，从而导致峰值流量偏低。

3.2. 二维正向溃口

二维正向模型

本节通过模拟 HR Wallingford 实验室为 IMPACT 项目（Morris & Hassan，2005）开展的两项二维正向溃坝试验，对侵蚀模型参数进行敏感性分析。所有试验均在一条长50 m、宽10 m 的水槽中进行；可冲刷坝体大致建于距水槽入口36 m 处（见图3a）。向水槽注水直至坝体上游水库蓄满，且水位超过坝体中部所开挖导流渠的高程，从而触发溃口开启。本文选取了标记为#2 和 #10 的两次试验，其主要试验特征汇总于表1。值得注意的是，试验#2 中坝体由无黏性材料（近乎均质的砂，d50 = 0.25 mm）筑成，而试验#10 则采用黏性材料（黏土）；因此，两次试验中的溃口演化过程存在差异。对于无黏性坝体，第一阶段与一维情形类似，表现为下游坡面均匀侵蚀，该坡面在对应导流渠的坝段处发生后退并趋于平缓；而对于黏性坝体试验，则在下游坡面观测到溯源侵蚀现象。此后，在两种情况下，溃口侧坡均开始失稳并发生岸坡坍塌，致使溃口随时间呈对称扩宽；当上游水库几近排空时，该过程结束。

500效果最佳



能确保模型准确预测坝顶被侵蚀后的高度变化。洪峰流量的预测尤为准确

图2. 一维正向溃坝试验：基于侵蚀模型对可蚀性系数 kd（单位：$cm^3·N^1·s^1$）开展的敏感性分析：（a）t = 30 s 和（b）t = 60 s 时刻的坝体剖面，以及（c）溢流流量。图（c）中同时给出了基于 Exner 方程模型预测的流量以供对比。虚线表示初始坝体剖面。

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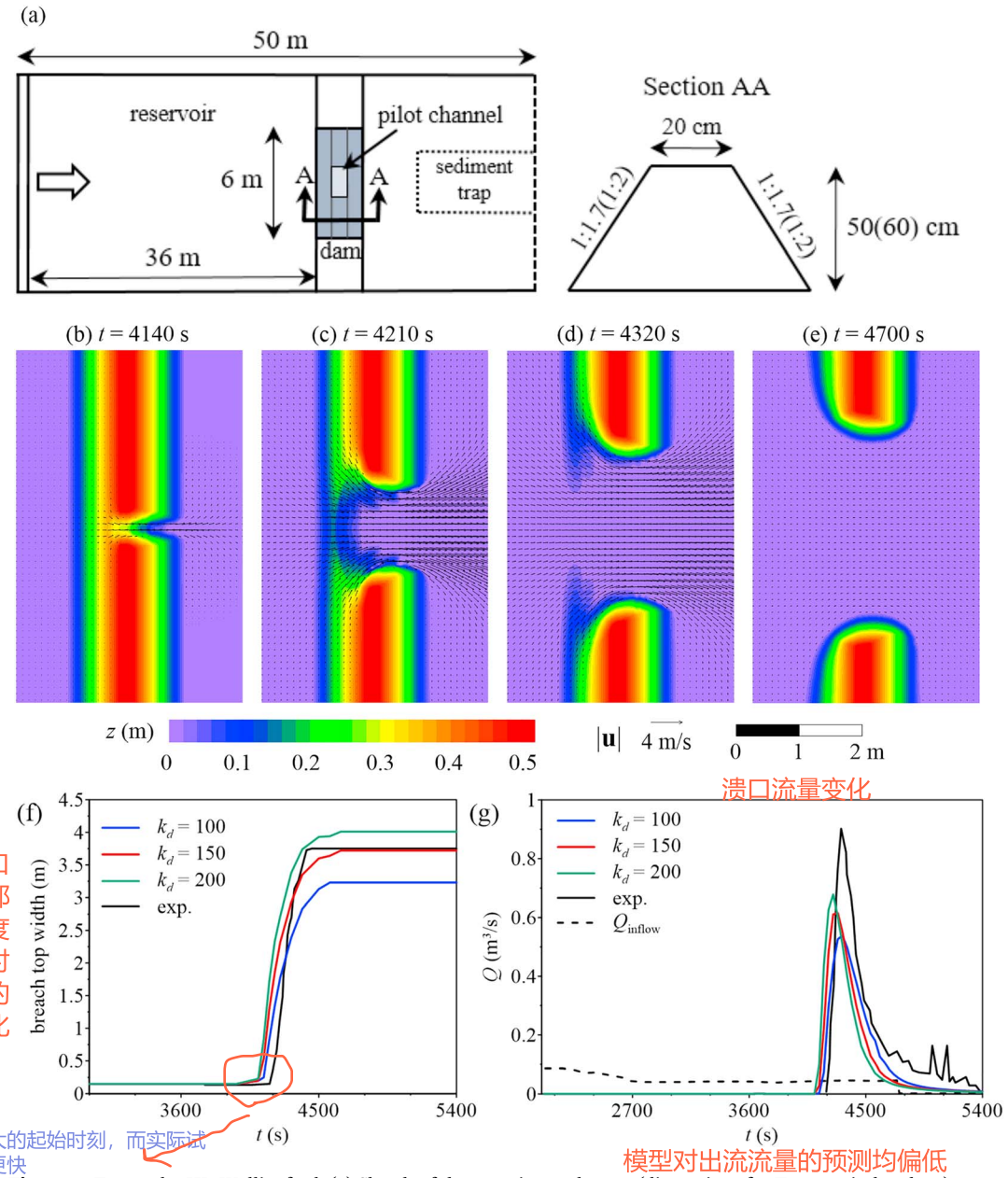


二维溃口随时间演变过程，以不同时刻的河床高程分布及流速矢量图表示

溃口流量变化

溃口顶部宽度随时间的变化

模型略微提前预测了溃口扩大的起始时刻，而实际试验中的扩展速度比数值模拟更快

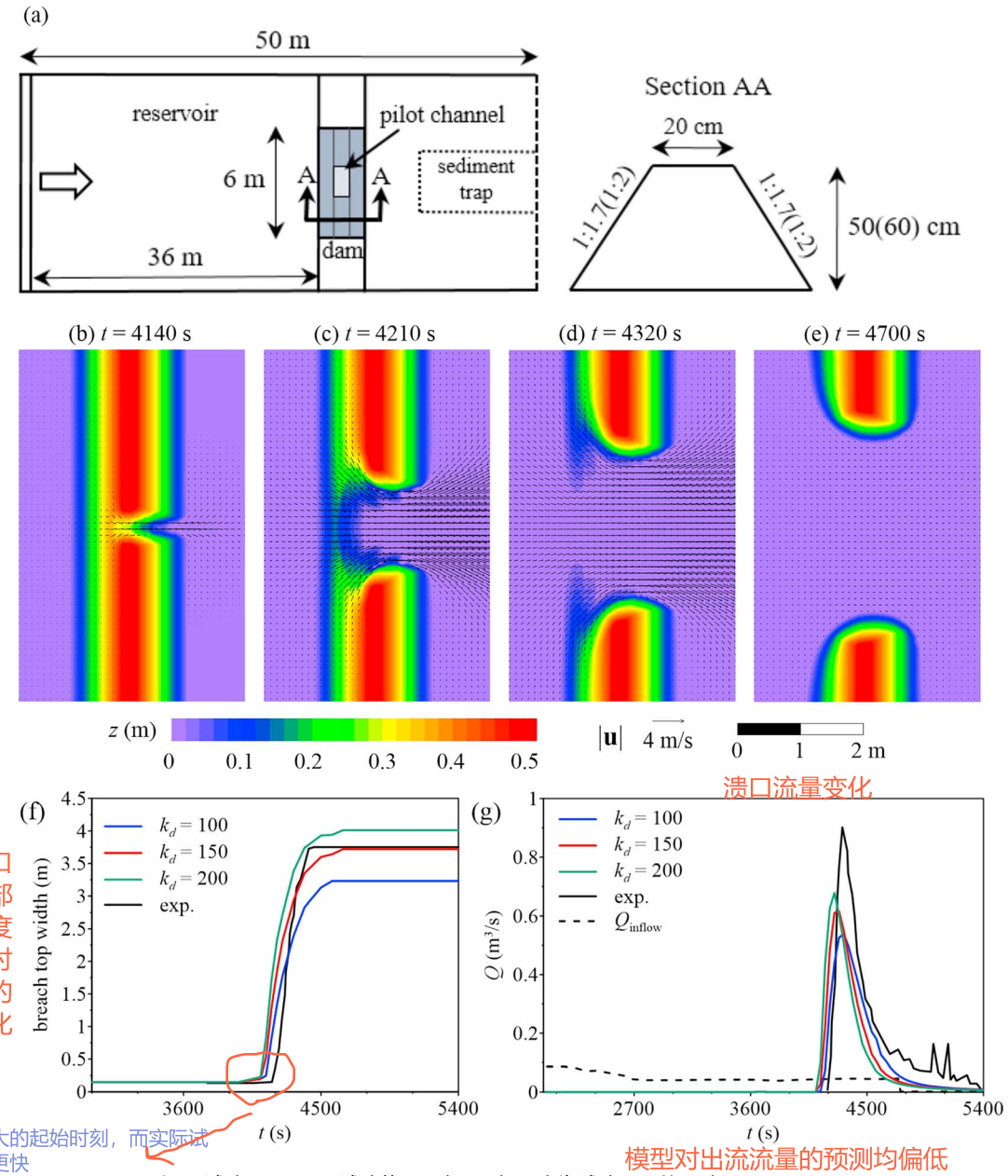
模型对出流流量的预测均偏低

Figure 3. Test #2 by HR Wallingford: (a) Sketch of the experimental setup (dimensions for Test #10 in brackets). (b–e) Simulated breach evolution in time, represented by the bottom elevation maps at selected times, and velocity vectors map. Experimental and numerical (f) breach top width evolution in time, and (g) discharge exiting through the breach; the inflow discharge is also represented in (g); numerical results are reported for three different values of the erodibility coefficient k_d (in $cm^3 \cdot N^{-1} \cdot s^{-1}$).

For both tests, the domain was discretized with square cells of size $\Delta x = \Delta y = 0.05$ m, and Manning's coefficient was set equal to 0.018 $m^{-1/3} \cdot s$ for Test #2 (as suggested by Wu et al., 2012) and to 0.016 $m^{-1/3} \cdot s$ for Test #10 (according to Wu, 2013). The time series for the inflow discharge were imposed as upstream boundary condition, while a free outlet condition was set downstream.

Test #2 is analyzed first. As in the previous test case, the values of the erodibility parameters were chosen in the range suggested by Hanson et al. (2010) for low clay content soils. In particular, the best fit was obtained with $\tau_c = 10^{-2}$ Pa, and $k_d = 150$ $cm^3 \cdot N^{-1} \cdot s^{-1}$. The other model parameters were set as follows: $p = 0.4$, $\varphi_{c,wet} = \varphi_{dep} = 30°$, and $\varphi_{c,dry} = 45°$. Figures 3b–3e show the bottom elevation contour maps at selected times, together with the velocity vectors. The absence of deposition downstream can be noticed,

二维溃口随时间演变过程，以不同时刻的河床高程分布及流速矢量图表示



溃口顶部宽度随时间的变化

溃口流量变化

模型略微提前预测了溃口扩大的起始时刻，而实际试验中的扩展速度比数值模拟更快

模型对出流流量的预测均偏低

图3. HR Wallingford 试验#2：（a）试验装置示意图（括号内为试验#10 的尺寸）。（b–e）模拟所得溃口随时间的演化过程，以若干典型时刻的河床高程图及流速矢量图表示；（f）实测与数值模拟所得溃口顶部宽度随时间的变化；（g）经溃口流出的流量；图（g）中亦标出了入流流量；数值结果对应三种不同可蚀性系数 kd（单位：$cm^3 \cdot N^1 \cdot s^1$）的取值。

两次试验中，计算域均采用尺寸为 Δx = Δy = 0.05 m 的正方形网格离散；曼宁系数在试验#2中设为 0.018 $m^{1/3} \cdot s$（参考 Wu 等人，2012），在试验#10中设为 0.016 $m^{1/3} \cdot s$（依据 Wu，2013）。上游边界条件采用给定的入流流量时序，下游边界则采用自由出流条件。

首先分析试验#2。与前述算例类似，可蚀性参数取值范围参照 Hanson 等人（2010）针对低黏土含量土壤所建议的区间。具体而言，最优拟合结果对应 τc = 10^2 Pa、kd = 150 $cm^3 \cdot N^1 \cdot s^1$。其余模型参数设定如下：p = 0.4，φc,wet = φdep = 30°，φc,dry = 45°。图3b–3e展示了若干典型时刻的河床高程等值线图，并叠加了流速矢量图。可观察到下游无泥沙淤积现象。

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Table 1
Main Features for the 2D Frontal Breach Test ases by HR Wallingford (Tests #2 and #10)

Feature	Test #2 非黏性	Test #10 黏性
Material type	Sand	Clay
d_{50} (mm)	0.25	0.005
Dam height (m)	0.50	0.60
Crest width (m)	0.20	0.20
Side slopes	1V:1.7H	1V:2H
Initial reservoir level (m)	0	0.58
Pilot channel depth (m)	0.02	0.05
Pilot channel width (m)	0.15	0.50

but the "hourglass" shape in the first phase of the breach opening and the final top width are well reproduced.

The measured and simulated breach top widths in time are reported in Figure 3f. Results obtained with three different values of the erodibility coefficient are compared (k_d = 100, 150, and 200 $cm^3 \cdot N^{-1} \cdot s^{-1}$, while all the other parameters are kept constant). The model with k_d = 150 $cm^3 \cdot N^{-1} \cdot s^{-1}$ is able to reproduce the breach evolution in time quite well. However, regardless of the selected erodibility coefficient, the model underestimates the outflow discharge, as can be observed in Figure 3g. The model slightly anticipates the beginning of the breach enlargement, which is more rapid in the experiments than in the numerical simulations, and this may influence the outflow discharge and the reservoir emptying process. Another possible cause of these discrepancies can be the uncertainty in the position and technique of the discharge measurements (the dam centerline is used as cross section for the discharge extraction in the numerical simulations, while no information is available as regards the experimental setup).

Additional sensitivity analyses were performed on Manning's coefficient, on the critical shear stress, and on the critical angles for slope stability, which are expected to influence the breach enlargement. The values 0.016 and 0.020 $m^{-1/3} \cdot s$ for Manning's coefficient were investigated. The critical shear stress τ_c was varied to 10^{-3} and 10^{-1} Pa, maintaining all the other parameters constant. The critical angle for slope stability in wet conditions $\varphi_{c,wet}$ (and the angle of deposition φ_{dep}) was changed from 30° to 25° and to 35°; as regards $\varphi_{c,dry}$, the values 40°, 50°, 60°, and 70° were examined. For all these simulations, the main breach characteristics (final top width and peak discharge) are reported in Table 2, and the relative error with reference to the experimental measurement and to the best fit simulation is computed. The critical shear stress has negligible influence on the results, as already noticed in the previous test case. Moreover, results are observed not to be much dependent on the "wet" critical angle for slope stability. On the other hand, an increase in the value of $\varphi_{c,dry}$ can reduce the final breach width and the peak discharge. Finally, Manning's coefficient does not particularly affect the main breach features. In fact, despite the fact that τ increases quadratically with n_f (equation (6)), at the same time velocity magnitude decreases, thus limiting the variation of the bed shear stress.

在模拟试验#10时，需设定适用于黏性土的冲蚀参数。

For simulating Test #10, erodibility parameters for a cohesive soil had to be set. Following Hanson et al. (2010), and also considering the formula for the estimation of k_d reported by Wu (2013), the range 0.1–10 $cm^3 \cdot N^{-1} \cdot s^{-1}$ can be considered adequate for the erodibility coefficient value of this kind of material. The critical shear stress τ_c should be set in the range 0.01–1 Pa. The following parameters were selected for the reference simulation: p = 0.4, τ_c = 0.1 Pa, k_d = 5 $cm^3 \cdot N^{-1} \cdot s^{-1}$, $\varphi_{c,wet} = \varphi_{dep}$ = 30°, and $\varphi_{c,dry}$ = 45°.

Table 2
Sensitivity Analysis on the Model Parameters for Test #2 by HR Wallingford 最终顶部宽度和峰值流量

Case	Peak discharge (m^3/s)	% error (measured)	% error (reference)	Final top width (m)	% error (measured)	% error (reference)
Measured	0.91	—	—	3.75	—	—
Reference simulation	0.63	−31	—	3.73	−1	—
k_d = 100 $cm^3 \cdot N^{-1} \cdot s^{-1}$	0.54	−41	−14	3.23	−14	−13
k_d = 200 $cm^3 \cdot N^{-1} \cdot s^{-1}$	0.68	−25	8	4.01	7	8
$\tau_c = 10^{-1}$ Pa	0.63	−31	0	3.73	−1	0
$\tau_c = 10^{-3}$ Pa	0.63	−31	0	3.73	−1	0
$\varphi_{c,wet} = \varphi_{dep}$ = 25°	0.64	−30	1	3.65	−3	−2
$\varphi_{c,wet} = \varphi_{dep}$ = 35°	0.62	−32	−2	3.7	−1	−1
$\varphi_{c,dry}$ = 40°	0.68	−25	8	4.15	11	11
$\varphi_{c,dry}$ = 50°	0.57	−37	−10	3.23	−14	−13
$\varphi_{c,dry}$ = 60°	0.49	−46	−22	2.39	−36	−36
$\varphi_{c,dry}$ = 70°	0.35	−62	−44	1.54	−59	−59
n_f = 0.016 $m^{-1/3} \cdot s$	0.60	−34	−5	3.64	−3	−2
n_f = 0.020 $m^{-1/3} \cdot s$	0.64	−30	2	3.81	2	2

Note. The reference simulation was performed adopting n_f = 0.018 $m^{-1/3} \cdot s$, $\tau_c = 10^{-2}$ Pa, k_d = 150 $cm^3 \cdot N^{-1} \cdot s^{-1}$, $\varphi_{c,wet} = \varphi_{dep}$ = 30°, and $\varphi_{c,dry}$ = 45°.

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表1
HR Wallingford 所开展的二维正向溃坝试验（试验#2 和 #10）主要特征

特征	试验 非黏性 #2	黏性 #10
材料类型	砂	黏土
d50（mm）		
坝高（m）		
坝顶宽度（m）		
边坡坡度		
初始水库水位（m）		
导流渠深度（m）		
导流渠宽度（m）		

但溃口开启初期的沙漏形形态及最终顶部宽度均得到了良好再现。

图3f展示了实测与模拟所得随时间变化的溃口顶部宽度。对比了三种不同可蚀性系数取值（kd = 100、150 和 200 $cm^3 \cdot N^1 \cdot s^1$）下的结果，其余参数均保持不变。kd = 150 $cm^3 \cdot N^1 \cdot s^1$ 的模型能较好地再现溃口随时间的演化过程。然而，无论选取何种可蚀性系数，该模型对出流量的预测均偏低，如图3g所示。模型略微提前预测了溃口扩宽的起始时刻；实验中溃口扩宽速率明显快于数值模拟，这可能影响出流量及水库排空过程。造成这些差异的另一可能原因是泄流量测量的位置与技术存在不确定性。

测量值（数值模拟中采用坝体中心线作为泄流提取的横断面，而实验装置的相关信息则未提供）。

还针对曼宁系数、临界剪切应力以及边坡稳定临界角开展了额外的敏感性分析，这些参数预计会影响溃口扩展过程。考察了曼宁系数的两个取值：0.016 和 0.020 m−1/3·s。临界剪切应力 τc 分别取为 10−3 Pa 和 10−1 Pa，其余参数保持不变。湿态下边坡稳定的临界角 φc,wet（即沉积角 φdep）分别设为 30°、25° 和 35°；对于干态临界角 φc,dry，则分别取 40°、50°、60° 和 70° 进行分析。所有这些模拟结果中的主要溃口特征（最终坝顶宽度和峰值泄流量）汇总于表2，并计算了相对于实测值及最优拟合模拟结果的相对误差。如前一试验案例中已观察到的那样，临界剪切应力对结果的影响可忽略不计。此外，结果亦显示对湿态边坡稳定临界角的依赖性不大。相反，增大 φc,dry 的取值可减小最终溃口宽度和峰值泄流量。最后，曼宁系数对主要溃口特征影响并不显著。事实上，尽管剪切应力 τ 随曼宁系数 nf 呈二次方增长（公式（6））， 但流速大小同时下降，从而限制了床面剪切应力的变化幅度。在模拟试验#10时，需设定黏性土的可侵蚀性参数。依据 Hanson 等（2010）的研究，并结合 Wu（2013）提出的 kd 估算公式，此类材料的可侵蚀系数 kd 取值范围 0.1–10 cm3·N−1·s−1 是适宜的；临界剪切应力 τc 应取值于 0.01–1 Pa 区间。参考模拟所采用的参数如下：孔隙率 p = 0.4，τc = 0.1 Pa，kd = 5 cm3·N−1·s−1，φc,wet = φdep = 30°，φc,dry = 45°。

在模拟试验#10时，需设定适用于黏性土的冲蚀参数。

表2
HR Wallingford 针对试验#2开展的模型参数敏感性分析

最终顶部宽度和峰值流量

工况	峰值泄流量（m^3/s）	实测误差（%）	参考模拟误差（%）	最终坝顶宽度（m）	实测误差（%）	参考模拟误差（%）
实测值		—	—	3.75	—	—
参考模拟		−31	—	3.73	−1	—
$k_d = 100\ cm^3 \cdot N^{-1} \cdot s^{-1}$	0.54	−41	−14	3.23	−14	−13
$k_d = 200\ cm^3 \cdot N^{-1} \cdot s^{-1}$	0.68	−25	8	4.01	7	8
$\tau_c = 10^{-1}$ Pa	0.63	−31	0	3.73	−1	0
$\tau_c = 10^{-3}$ Pa	0.63	−31	0	3.73	−1	0
$\varphi_{c,wet} = \varphi_{dep} = 25°$	0.64	−30	1	3.65	−3	−2
$\varphi_{c,wet} = \varphi_{dep} = 35°$	0.62	−32	−2	3.7	−1	−1
$\varphi_{c,dry} = 40°$	0.68	−25	8	4.15	11	11
$\varphi_{c,dry} = 50°$	0.57	−37	−10	3.23	−14	−13
$\varphi_{c,dry} = 60°$	0.49	−46	−22	2.39	−36	−36
$\varphi_{c,dry} = 70°$	0.35	−62	−44	1.54	−59	−59
nf = 0.016 $m^{1/3} \cdot s$		−34	−5	3.64	−3	−2
nf = 0.020 $m^{1/3} \cdot s$		−30	2	3.81	2	2

注：基准模拟采用以下参数：nf = 0.018 $m^{1/3} \cdot s$，τc = 10^2 Pa，kd = 150 $cm^3 \cdot N^1 \cdot s^1$，φc,wet = φdep = 30°，φc,dry = 45°。

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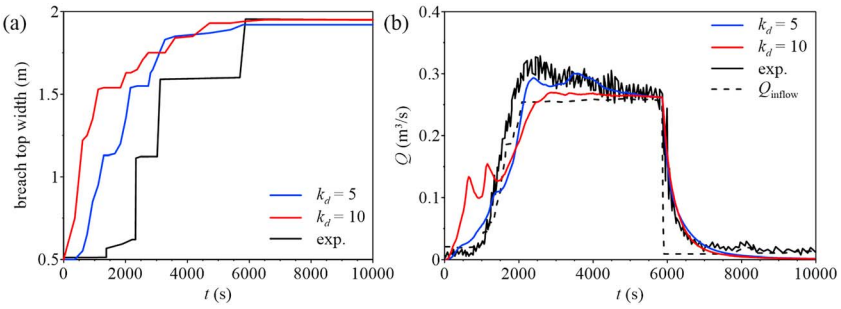
由于浅水方程（SWE）假设的局限性，模型无法捕捉实验中观察到的溯源侵蚀现象



因溃口迅速发展而导致初期流量骤增，且峰值略偏低。

Figure 4. Test #10 by HR Wallingford. Experimental and numerical (a) breach top width evolution in time, and (b) discharge exiting through the breach. The inflow discharge is also represented. Numerical results are reported for two different values of the erodibility coefficient k_d (in $cm^3 \cdot N^{-1} \cdot s^{-1}$).

The dynamics of the breach evolution predicted by the model, though much slower because of the reduced erodibility of the dam material, is similar to Test #2. Due to the limitation of the SWE assumptions, the model is not able to predict the headcut erosion observed experimentally; in spite of this, the final top width is well reproduced (Figure 4a), even if the enlargement process is faster than in the measurements. In this case, since the dynamics is much slower than in the case of Test #2, this discrepancy does not influence the reservoir emptying very much. In fact, the outflow discharge fits the one registered during the experiments well (Figure 4b), in both shape and peak discharge.

When values of the erodibility coefficient smaller than 5 $cm^3 \cdot N^{-1} \cdot s^{-1}$ were adopted, the breach formation process was initially too slow, and the increase in the water level upstream caused the overtopping of the whole dam (not limited to the pilot channel) and the consequent widespread erosion, which was never observed in the experiments. This probably happens because the initial evolution of the breach is generated by headcut erosion, which cannot be simulated with SWE models. Hence, only one larger value of k_d was considered in the sensitivity analysis ($k_d = 10$ $cm^3 \cdot N^{-1} \cdot s^{-1}$), and numerical results for this simulation are compared with the experiments and with the reference simulation in Figure 4. The breach opening is faster than in the reference simulation, but the final top width is still well caught; the overflow discharge presents a slightly different trend, with a sudden initial increase due to the rapid breach erosion and an underestimated peak value. Also for this test case, a sensitivity analysis on the critical shear stress revealed that this parameter does not influence the model predictions (results not shown).

3.3. Experimental Levee Breach

堤防侧向溃口试验（水流垂直于溃口!）
检验所采用的侵蚀模型能否合理再现堤防侧向溃口的开启过程

The aim of the present test case is to investigate how the erosion model can reproduce the opening of lateral breaches. In particular, one of the field-scale levee breach experimental tests presented by Kakinuma and Shimizu (2014) is considered (Case 4). A 176-m-long and 8-m-wide stretch of the floodway channel of the Tokachi River (Japan), with bottom slope equal to 1/500, was set up by inserting a vertical wall on the left and by substituting a portion of the existing right levee with a 3-m-high erodible dyke, made of sand with $d_{50} = 0.7$ mm, $d_{30} \approx 0.2$ mm, $d_{90} \approx 40$ mm, $\rho_s = 2{,}650$ kg/m^3, and $p = 0.4$. The levee crest width was equal to 6 m, while the side slopes were both 1V:2H. The inflow discharge was increased until the levee was overtopped just at the location where a notch (with length 3 m and depth 0.5 m) had been previously carved to trigger the breach; the outflow discharge inundated a floodable area specifically arranged (Figure 5a).

The grid size was set at 0.5 m, while Manning's coefficient was assumed equal to 0.023 $m^{-1/3} \cdot s$, as suggested by the experimenters. The measured inflow discharge was set as upstream boundary condition, and a rating curve was imposed downstream far enough to avoid disturbances in the water level at the breach site. The following parameters were assumed for the levee material for the erosion model: $\tau_c = 0.5$ Pa, $k_d = 80$ $cm^3 \cdot N^{-1} \cdot s^{-1}$, $\varphi_{c,wet} = \varphi_{dep} = 30°$, and $\varphi_{c,dry} = 40°$. The same critical angles were used for the sediment transport model.

Experimental observations show that, initially, overflow water starts eroding the downhill slope of the levee, until the top of the front slope is reached, and erosion proceeds downward to the bottom of the levee; then, the breach begins to widen to both sides. However, soon, the breach is observed to widen

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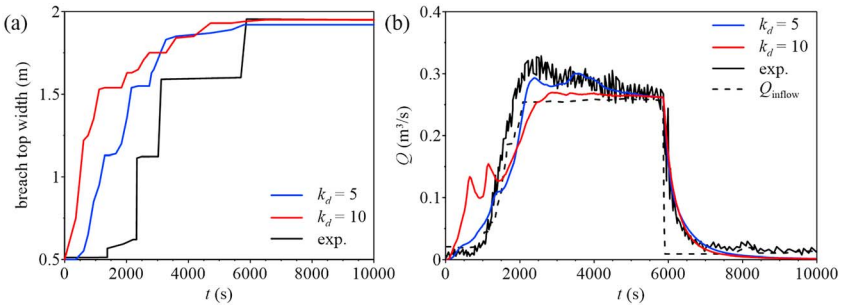
由于浅水方程（SWE）假设的局限性，模型无法捕捉实验中观察到的溯源侵蚀现象



因溃口迅速发展而导致初期流量骤增，且峰值略偏低。

图4. HR Wallingford公司试验#10。实验与数值模拟结果：（a）溃口顶部宽度随时间变化，（b）经溃口流出的流量。图中同时给出入流流量。数值结果对应两个不同可侵蚀性系数kd值（单位：cm³·N¹·s¹）。

模型预测的溃口演化动力学过程虽因坝体材料可侵蚀性降低而显著变缓，但整体形态与试验#2相似。受限于浅水方程（SWE）假设，模型无法预测实验中观测到的溯源侵蚀现象；尽管如此，最终顶部宽度仍得到良好再现（图4a），仅扩展过程较实测更快。本案例中，由于溃口演化动力学过程远比试验#2缓慢，上述差异对水库排空过程影响甚微。事实上，出流流量过程线在形状与峰值流量两方面均与实验实测结果高度吻合（图4b）。

当可侵蚀性系数取值小于5 cm³·N¹·s¹时，溃口形成过程初始阶段过于缓慢，导致上游水位持续升高，最终致使整个坝体（而非仅限于导流渠）发生漫顶，并引发大范围侵蚀，这在实验中从未出现。此现象很可能源于溃口初始演化由溯源侵蚀主导，而浅水方程（SWE）模型无法模拟此类过程。因此，敏感性分析中仅考虑一个较大的kd值（kd = 10 cm³·N¹·s¹），并将该模拟结果与实验及基准模拟结果一并在图4中对比。该模拟下溃口开启速度更快，但最终顶部宽度仍得到较好再现；溢流流量过程线呈现略有不同的趋势：因溃口快速侵蚀导致初始流量骤增，且峰值流量被低估。对于本测试案例，临界剪切应力的敏感性分析同样表明该参数对模型预测结果无显著影响（结果未展示）。

3.3 实验堤防溃口

堤防侧向溃口试验（水流垂直于溃口!）
检验所采用的侵蚀模型能否合理再现堤防侧向溃口的开启过程

本测试案例旨在考察侵蚀模型对侧向溃口开启过程的模拟能力。具体选取Kakinuma与Shimizu（2014）提出的野外尺度堤防溃口试验之一（案例4）作为研究对象。试验在十胜川（日本）行洪道一段长176米、宽8米的渠段上开展，渠底坡度为1/500；左侧设置垂直挡墙，右侧原有堤防的一部分被替换为一座高3米、由砂土构成的可侵蚀堤坝（d = 0.7 mm，d ≈ 0.2 mm，d ≈ 40 mm，ρ = 2650 kg/m³，孔隙率p = 0.4）。堤顶宽6米，两侧边坡均为1V:2H。通过逐步增大入流流量，使水流恰好在预先开凿的缺口（长3米、深0.5米）处漫过堤防以触发溃口；出流淹没专门布置的可淹没区域（图5a）。网格尺寸设定为0.5米，曼宁糙率系数取试验者建议值0.023 m¹/³·s。上游边界条件采用实测入流流量，下游边界采用足够远距离的水位—流量关系曲线，以避免对溃口处水位造成干扰。侵蚀模型中堤防材料参数设定如下：τc = 0.5 Pa，kd = 80 cm³·N¹·s¹，湿态临界休止角φc,wet = 淤积角φdep = 30°，干态临界休止角φc,dry = 40°。泥沙输运模型采用相同临界休止角。

实验观测显示，初始阶段漫溢水流首先冲刷堤防下游坡面，直至抵达前坡顶部，继而向下侵蚀至堤脚；此后溃口开始向两侧扩展。然而很快即可观察到溃口向一侧扩展速率明显快于另一侧。

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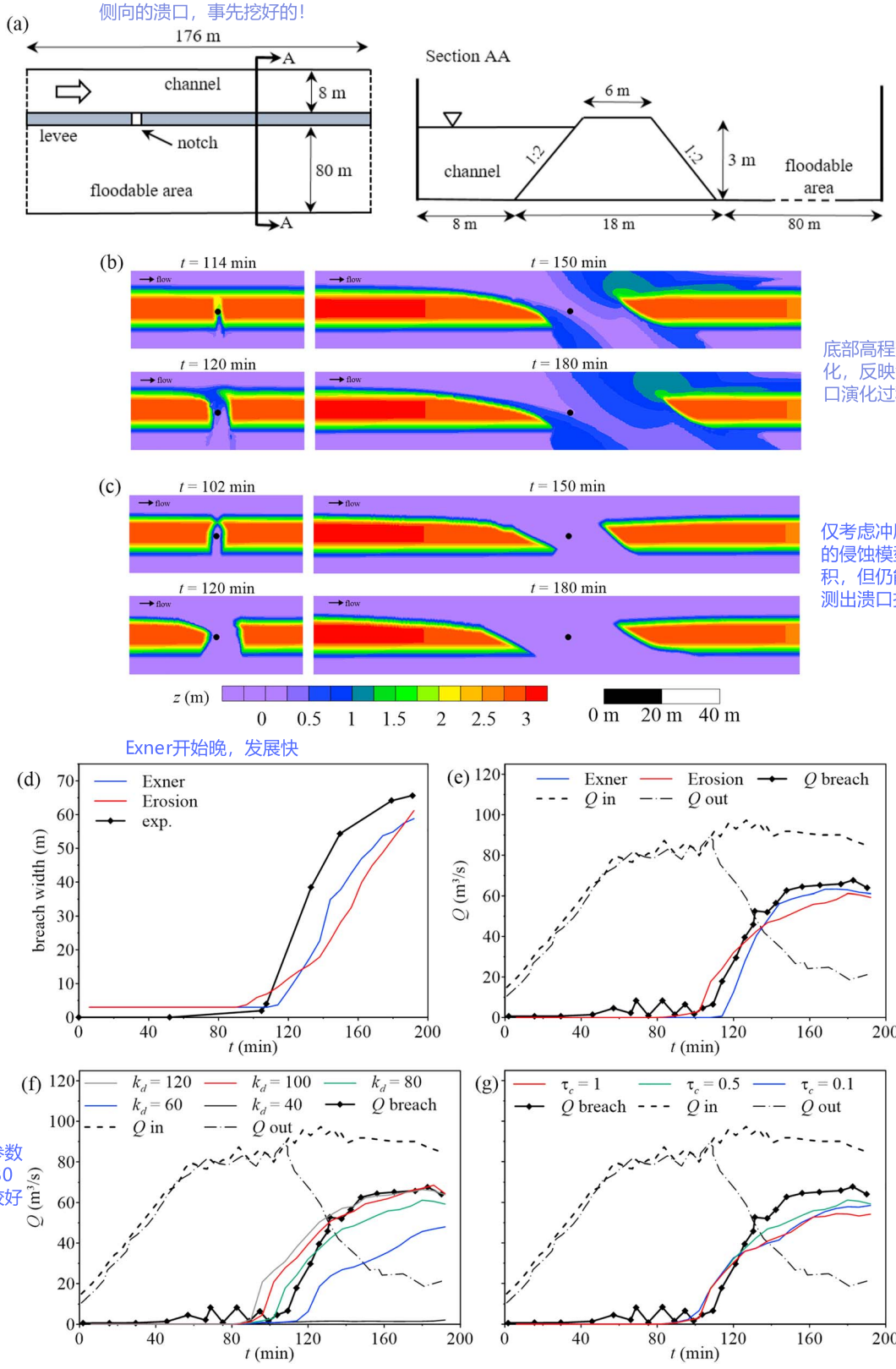
侧向的溃口，事先挖好的！



底部高程变化，反映溃口演化过程

仅考虑冲刷而忽略沉积作用的侵蚀模型无法再现泥沙淤积，但仍能在一定程度上预测出溃口扩展的不对称性

Exner开始晚，发展快

可蚀性参数Kd大于80效果比较好

洪峰到达时间和峰值流量对临界剪切应力的变化并不敏感。

Figure 5. Levee breach experiment. (a) Sketch of the test setup. Breach evolution in time, represented by bottom elevation maps at selected times, as predicted by (b) the sediment transport model and by (c) the erosion model. (d) Breach width evolution in time (experimental and numerical). (e) Discharge exiting through the breach (experimental and numerical). Discharge exiting through the breach: Results of the sensitivity analysis on (f) the erodibility coefficient k_d (in $cm^3/N/s$) and on (g) the critical shear stress τ_c (in Pa). The inflow and outflow discharge time series in the channel are also reported in panels (e–g).

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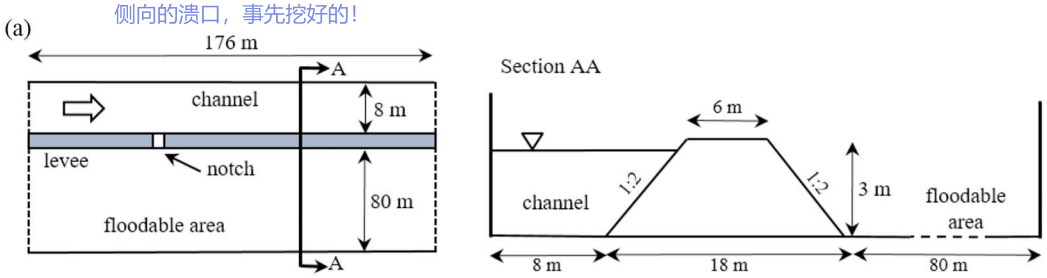
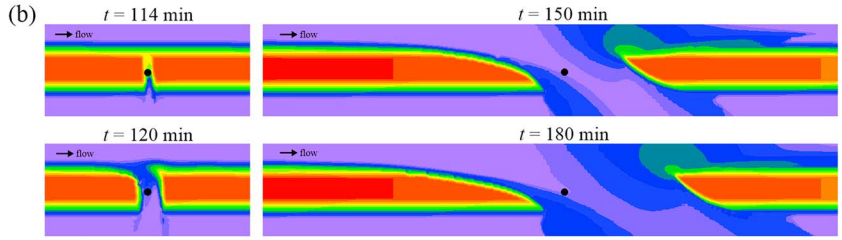
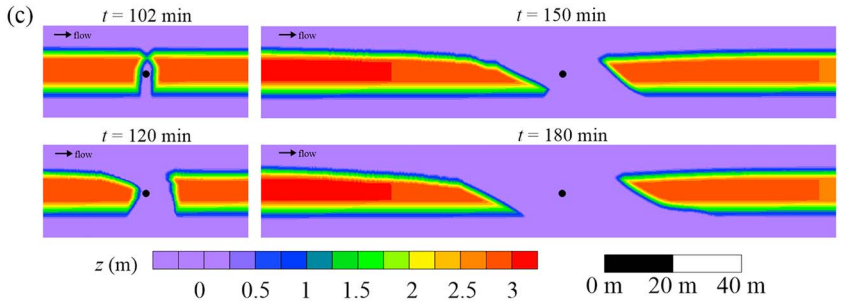
侧向的溃口，事先挖好的！



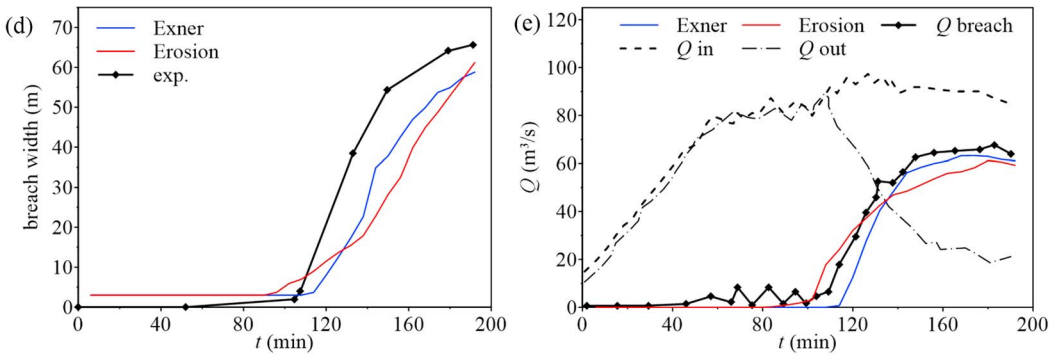


底部高程变化，反映溃口演化过程

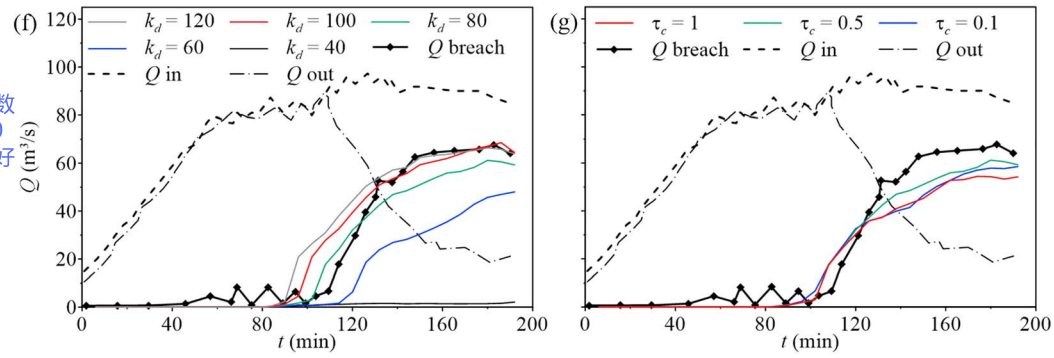


仅考虑冲刷而忽略沉积作用的侵蚀模型无法再现泥沙淤积，但仍能在一定程度上预测出溃口扩展的不对称性

Exner开始晚，发展快



可蚀性参数Kd大于80效果比较好



洪峰到达时间和峰值流量对临界剪切应力的变化并不敏感。

图5. 堤防溃口试验。（a）试验装置示意图。（b）泥沙输运模型与（c）侵蚀模型预测的溃口随时间演化过程（以选定时刻床面高程图表示）。（d）溃口宽度随时间变化（实验与数值结果）。（e）经溃口流出的流量（实验与数值结果）。经溃口流出的流量：（f）可侵蚀性系数kd（单位：cm³/N/s）与（g）临界剪切应力τc（单位：Pa）敏感性分析结果。各子图（e–g）中同时给出了渠道内入流与出流流量过程线。

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描述了横向溃口过程中，溃口不对称的原因，泥沙输移模型能够较好地捕捉这一过程

at a much higher rate in the downstream direction than in the upstream direction; this is due to the development of a high-velocity flow band near the downstream end of the breach, and of a dead water area near the upstream end, where sedimentation occurs. The asymmetry of the breach final width with reference to the initial notch position is a typical feature of lateral breaches, in contrast with what is usually observed in frontal dam breach test cases. The sediment transport model is able to capture this process, as can be noticed in Figure 5b, where the bed elevation contour maps at selected times are reported to show the breach evolution. On the other hand, the erosion model, which neglects deposition, is not able to reproduce this process in detail; nevertheless, the asymmetry of the breach widening is still predicted, especially in the first stages (see Figure 5c), even if in a less pronounced way than in the experimental observations toward the end of the process. Despite these differences in the simulation of the erosion process, the two models predict a similar trend for the total breach width, which is slightly underestimated with reference to the experimental data, as can be noticed in Figure 5d. Breach widening seems to start somewhat late in the sediment transport model simulation but then evolves at a higher rate than in the erosion model simulation. A similar trend can be observed in the outflow discharge time series, reported in Figure 5e. The peak discharge is underestimated by only 4–5% by both models with reference to the measured value.

A sensitivity analysis on the erodibility parameters was performed. First, the erodibility coefficient was analyzed, and simulations were repeated assuming $k_d = 40$, 60, 80, 100, and 120 $cm^3 \cdot N^{-1} \cdot s^{-1}$ ($\tau_c = 0.5$ Pa). The final breach width is observed to increase with erodibility (simulated values are 8, 33, 62, 80, and 127 m, respectively). The smallest erodibility value is probably not representative of the levee material, since overtopping does not generate appreciable erosion; on the other hand, the highest erodibility values overestimate the final breach width, also because erosion is predicted along the inner riverbank (upstream of the breach), and this fact was not observed experimentally. As regards the predicted discharges, Figure 5f compares the outflow hydrographs: The arrival time is slightly anticipated for the highest erodibility values, but the final peak discharge is very similar for $k_d \geq 80$ $cm^3 \cdot N^{-1} \cdot s^{-1}$.

The sensitivity to the critical shear stress was also analyzed, and τ_c was varied from 0.1 to 0.5 and 1 Pa (maintaining $k_d = 80$ $cm^3 \cdot N^{-1} \cdot s^{-1}$). The outflow hydrographs obtained from the three simulations are compared in Figure 5g, showing that the arrival time and the peak discharge are only slightly dependent on the critical shear stress.

3.4. Levee Breach on the Enza River 埃恩扎河堤防溃口案例，，以验证其在真实案例中的适用性

The model was finally employed to simulate the recent flood event that took place on the Enza River (Northern Italy), a tributary of the Po River, in order to verify its applicability to real test cases. A severe flood event followed the prolonged heavy rainfall occurred on the river basin on 10–11 December 2017, resulting in the highest water levels ever recorded at all the gauging stations along the river. On 12 December at 05:30 a.m. water started to overtop the right levee near Lentigione di Brescello (Reggio Emilia), initially triggering three very close breaches, which almost merged into a single large one in time. The overtopped part of the levee was 250 m long, and the total final breach width was approximately 160 m, while the widening took about 4 hr. The total flooded area was about 6.3 km^2, restricted by the levees of the Enza and Po Rivers, by a road embankment and a channel levee (see Figure 6a).

The terrain elevation was obtained from a digital terrain model with a resolution equal to 1 m, based on a LiDAR survey of the area. The domain was then discretized with square cells of size 2 m × 2 m (approximately 5 million active cells), but the levee crest elevations were preserved after the downsampling of the original digital terrain model. The roughness coefficient was set equal to 0.05 $m^{-1/3} \cdot s$, after a calibration procedure. The upstream boundary condition is the discharge time series obtained from the conversion of measured water levels at the level gauge station of Sorbolo (whose position is reported in Figure 6a). An experimental rating curve was available, but the presence of the breach is expected to influence the level measurements due to the generation of a drawdown profile, thus “invalidating” the rating curve after the breach opening. For this reason, two different stage-discharge relations (before and after the breach) were used to convert the water levels, following the same procedure described by Vacondio et al. (2016). The discharge hydrograph is shown in Figure 6b, together with the water levels in the Po River, imposed downstream (these were relatively low, and backwater effects were not observed during the event).

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描述了横向溃口过程中，溃口不对称的原因，泥沙输移模型能够较好地捕捉这一过程

溃口向下游方向的扩展速率远高于上游方向；这是由于溃口下游端附近形成高速水流带，而上游端附近则形成滞水区并发生泥沙淤积。最终溃口宽度相对于初始缺口位置呈现不对称分布，这是侧向溃口的典型特征，与通常在正向坝体溃口试验中观察到的现象截然不同。泥沙输运模型能够准确捕捉该过程，如图5b所示——其中展示了选定时刻床面高程等值线图，用以反映溃口演化过程。另一方面，忽略淤积作用的侵蚀模型虽无法细致再现该过程，但仍能预测出溃口扩展的不对称性，尤其在初期阶段（见图5c），尽管其程度较实验观测结果在过程末期所呈现的弱一些。尽管两种模型在侵蚀过程模拟上存在上述差异，但它们对总溃口宽度的演化趋势预测基本一致，且均略低于实验数据（见图5d）。泥沙输运模型模拟中溃口扩展启动稍晚，但随后扩展速率高于侵蚀模型模拟。图5e所示出流流量过程线也呈现类似趋势。两种模型对峰值流量的低估幅度仅为实测值的4–5%。

开展了针对可侵蚀性参数的敏感性分析。首先分析了可侵蚀性系数，分别取kd = 40、60、80、100和120 $cm^3 \cdot N^1 \cdot s^1$（τc = 0.5 Pa）重复进行模拟。结果显示，最终溃口宽度随可侵蚀性增大而增加（对应模拟值分别为8、33、62、80和127米）。最小可侵蚀性值可能无法代表实际堤防材料特性，因为此时漫顶几乎不引发明显侵蚀；而最大可侵蚀性值则高估了最终溃口宽度，部分原因在于模型预测溃口上游内侧河岸亦发生侵蚀，但实验中并未观测到该现象。关于预测流量，图5f对比了各次模拟的出流过程线：最大可侵蚀性条件下洪峰到达时间略有提前，但当kd ≥ 80 $cm^3 \cdot N^1 \cdot s^1$时，最终峰值流量十分接近。

还分析了模型对临界剪切应力的敏感性，将τc分别设为0.1、0.5和1 Pa（保持kd = 80 $cm^3 \cdot N^1 \cdot s^1$）。图5g对比了三次模拟所得出流过程线，结果表明洪峰到达时间与峰值流量仅轻微依赖于临界剪切应力。

3.4 恩扎河堤防溃口

埃恩扎河堤防溃口案例，，以验证其在真实案例中的适用性

该模型最终被用于模拟近期发生在意大利北部恩扎河（波河支流）的洪水事件，以验证其在真实测试案例中的适用性。2017年12月10日至11日，流域内持续强降雨引发严重洪水，导致沿河所有水文站均记录到有史以来最高水位。12月12日清晨5时30分，水体开始漫过布雷斯切洛伦蒂焦内（雷焦艾米利亚省）附近右岸堤防，最初引发三处位置极为接近的溃口，随后逐渐合并为一处大型溃口。漫顶段堤防长度为250米，最终总溃口宽度约160米，溃口扩展过程历时约4小时。受恩扎河与波河堤防、一条公路路基及一条渠道堤防所限（见图6a），总淹没面积约6.3平方公里。

地形高程数据源自基于该区域激光雷达（LiDAR）测量的数字地形模型，分辨率为1米。随后，计算域采用2米×2米的正方形网格离散化（约500万个活跃网格），但下采样原始数字地形模型后仍保留堤顶高程。糙率系数经率定后设为0.05 $m^{1/3} \cdot s$。上游边界条件为索尔博洛水位站实测水位经换算所得的流量过程线（其位置见图6a）。虽有现成的经验水位—流量关系曲线，但溃口形成后产生的落水剖面将影响水位测量，致使溃口开启后该关系曲线失效。因此，参照Vacondio等（2016）所述方法，采用溃口开启前后两套不同的水位—流量关系进行水位换算。图6b展示了流量过程线，以及下游波河所施加的水位（水位相对较低，事件期间未观测到回水影响）。

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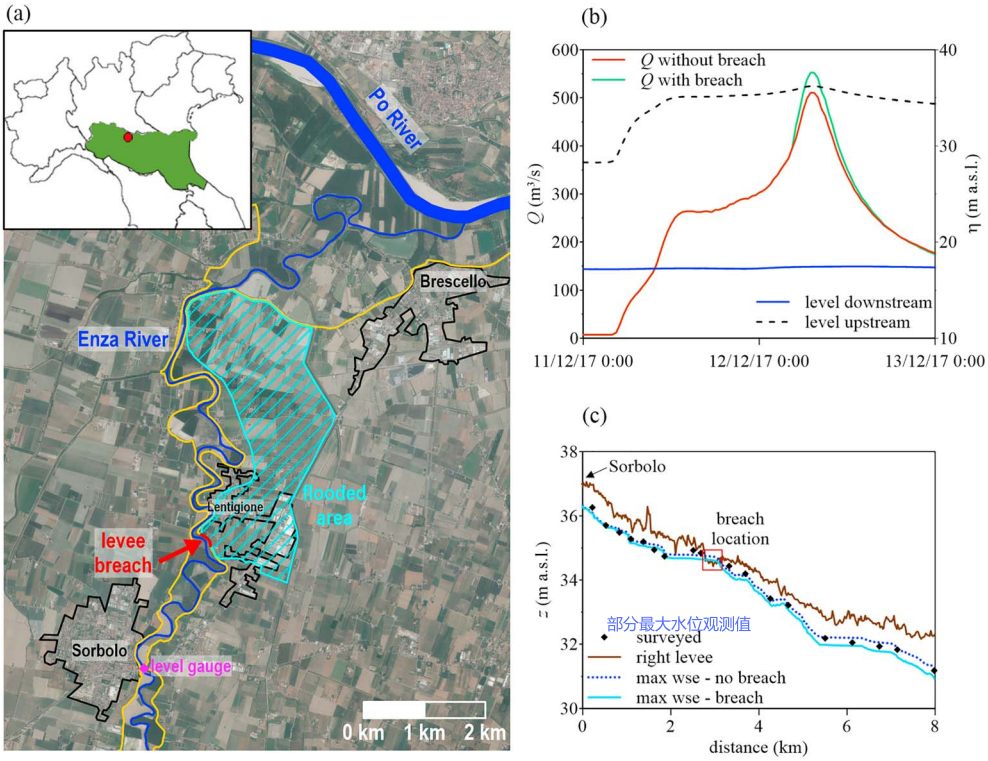

Figure 6. Levee breach on the Enza River. (a) Sketch of the study area: The Enza and Po Rivers (blue), the levees (yellow), the breach location (red), the main residential and industrial areas (black), and the flooded area (cyan) are identified on an orthoimage. (b) Discharge time series imposed as upstream boundary condition and water levels imposed downstream. (c) Longitudinal profile of the (right) levee crest elevation and of the maximum water surface elevations obtained from simulations with and without the breach erosion. The maximum surveyed water levels at selected locations are also reported for comparison.

所选堤防材料参数

The levee is built with silt loam with the following texture: sand 15–39%, silt 49–67%, and clay 12–18%. The material porosity is equal to 0.4, and the dry specific weight of soil is approximately 1.55 Mg/m^3. Considering these characteristics, Hanson et al. (2010) suggest an erodibility coefficient in the range 0.5–10 cm^3·N^{-1}·s^{-1}, while the critical shear stress should be assumed in the range 0.1–1 Pa. Moreover, the formula reported by Wu (2013) for estimating k_d would lead to 2–3 cm^3·N^{-1}·s^{-1}. The following parameters were then assumed for the levee material for the erosion model: τ_c = 1 Pa, k_d = 5 cm^3·N^{-1}·s^{-1}, p = 0.4, $\varphi_{c,wet} = \varphi_{dep}$ = 30°, and $\varphi_{c,dry}$ = 50°. The erosion equation was applied only in the cells representing the levee (hence, the riverbed is never modified), and the bottom elevation of the levee foundation (assumed equal to the local terrain elevation outside the river) was also specified, in order to prevent erosion below the ground level. Note that erosion can potentially occur anywhere on the levees; the exact position of the breach does not have to be defined a priori, because the opening occurs where the levee is overtopped. Figure 6c reports the longitudinal profile of the (right) levee crest elevation for the 8-km-long stretch of the river downstream of Sorbolo, together with the profiles of the maximum water surface elevations along the river obtained from two different simulations. In the first one the levees are assumed to be nonerodible (bathymetry constant in time in the whole domain), whereas in the second one the bathymetry can change accordingly to equation (7) and to the bank failure algorithm, leading to the breach formation where the levee is overtopped. In the second simulation, the maximum water levels are lowered due either to the drawdown effect induced by the breach upstream or to the reduced discharge downstream. The maximum observed water levels surveyed after the event are also reported at selected locations and confirm that the flood propagation along the river is correctly reproduced by the model (Figure 6c). Please note that the water levels (both simulated and surveyed ones) are quite close to the levee crest in different locations, but the levee is actually overtopped only where breaches were observed in the field.

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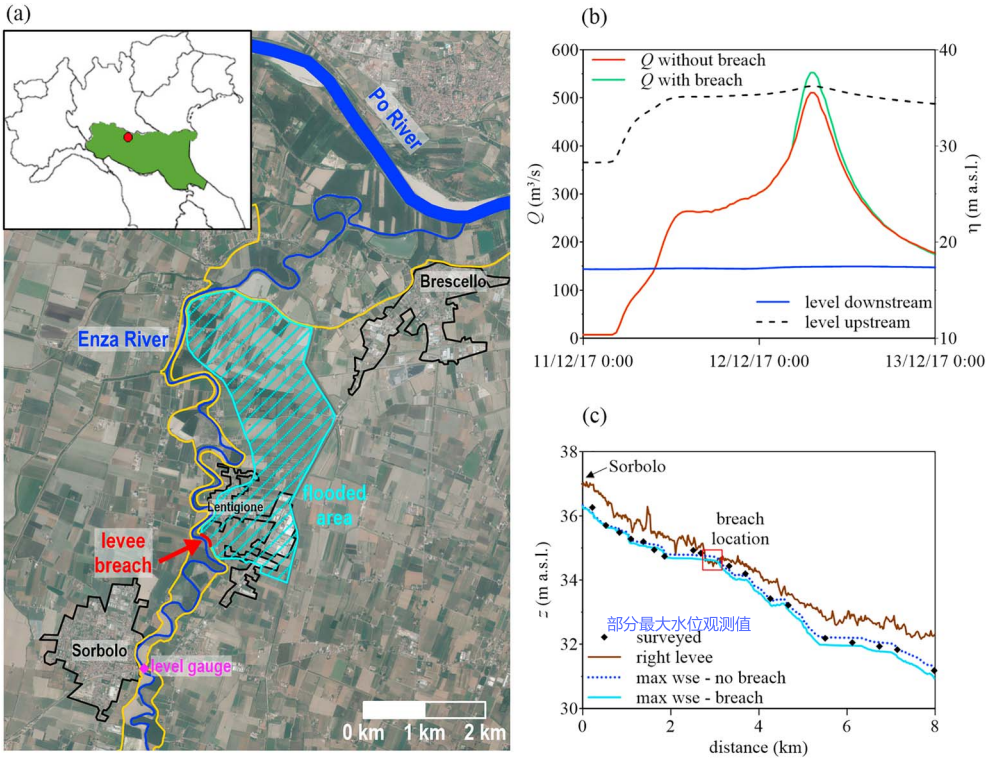

图6. 恩扎河堤防溃口。（a）研究区域示意图：恩扎河与波河（蓝色）、堤防（黄色）、溃口位置（红色）、主要居民与工业区（黑色）及淹没区域（青色）在正射影像上标出。（b）上游边界条件所施加的流量过程线及下游所施加的水位。（c）（右岸）堤顶高程纵向剖面及含溃口侵蚀与不含溃口侵蚀两种模拟所得最大水面高程剖面。同时标出若干位置实测的最大水位用于对比。

所选堤防材料参数

该堤防由粉质壤土筑成，其颗粒组成如下：砂粒15–39%，粉粒49–67%，黏粒12–18%。材料孔隙率为0.4，土壤干容重约为1.55 Mg/m³。依据这些特性，Hanson等（2010）建议可蚀性系数取值范围为0.5–10 $cm^3 \cdot N^1 \cdot s^1$，临界剪应力取值范围为0.1–1 Pa。此外，Wu（2013）提出的kd估算公式给出的结果为2–3 $cm^3 \cdot N^1 \cdot s^1$。据此，溃蚀模型中堤防材料所采用的参数如下：τc = 1 Pa，kd = 5 $cm^3 \cdot N^1 \cdot s^1$，p = 0.4，φc,wet = φdep = 30°，φc,dry = 50°。溃蚀方程仅应用于代表堤防的网格单元（因此河床高程永不改变），同时指定了堤基底部高程（假定等于堤外局部地形高程），以防止侵蚀低于地面高程。需注意，侵蚀可能发生在堤防任意位置；溃口确切位置无需预先设定，因溃口将在堤防发生漫溢处自然形成。图6c给出了索尔博洛下游8公里河段（右岸）堤顶高程的纵向剖面，以及两次不同模拟所得河道内最大水面高程剖面。第一次模拟中假设堤防不可侵蚀（全流域地形高程恒定），第二次模拟则允许地形高程按式（7）及岸坡失稳算法动态变化，从而在堤防漫溢处形成溃口。第二次模拟中，最大水位降低，原因或是上游溃口引起的落水效应，或是下游流量减小所致。事件后实测的最大水位亦在若干位置标出，证实模型对洪水沿河道传播过程的再现是准确的（图6c）。请注意，实测与模拟水位在不同位置均十分接近堤顶高程，但堤防实际仅在野外观测到溃口的位置发生漫溢。

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该区域堤顶高程并不均匀，某些低洼处更容易受到冲刷，从而形成多个小型溃口，并随时间逐渐扩大。

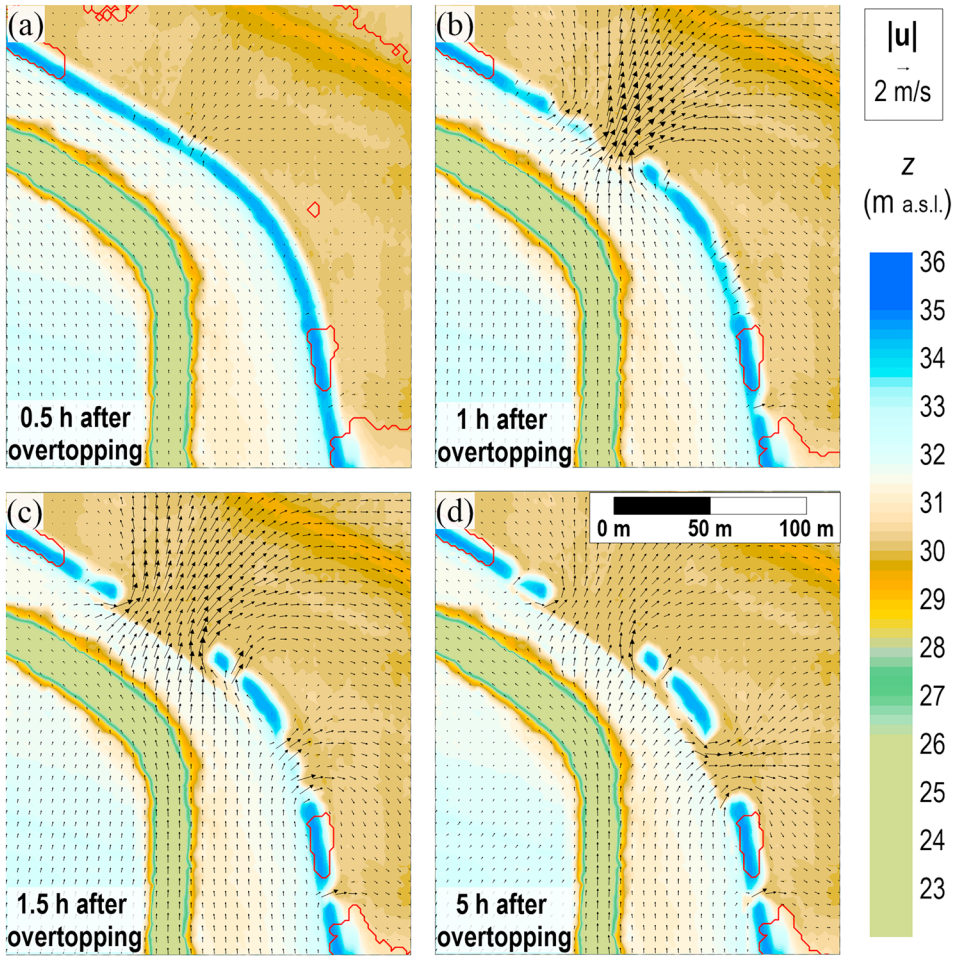

Figure 7. Levee breach on the Enza River. Contour maps of the simulated bottom elevation and vector maps of the velocity in the breach zone at selected times: (a) 0.5 hr, (b) 1 hr, (c) 1.5 hr, and (d) 5 hr after the beginning of overtopping. Only one vector out of nine is represented for the sake of clarity. The red contour line identifies the wet/dry front.

Figure 7 shows the breach evolution in terms of contour maps of the bottom elevation at selected times (velocity vectors are also reported). Initially, erosion is not concentrated in a restricted area (as in the previous test case), because water overtops the levee crest along roughly 270 m, and the erosion process appears scattered along this length. However, the levee crest elevation is not regular in this area; hence, some low points appear more vulnerable to erosion and give origin to the development of multiple small breaches, which enlarge in time. The most vulnerable point is just downstream of the levee bend, and the highest velocities are observed there. The breach takes about 3 hr from the beginning of overtopping to reach its final extension, with only small further modifications in the following 2 hr. The total final width is approximately 150 m, separated into five segments along roughly 250 m, the largest of which is 60 m wide. Actually, only three breaches were observed in the field; however, given the uncertainties in boundary conditions, material parameters, and terrain elevations, the model correctly predicts the opening of multiple breaches and captures the overall process quite well. Clearly, any model that only employs geometric relations or internal links between river and floodplain to simulate the breach opening would hardly capture this complex behavior.

The actual total flooded area is reported in Figure 8, compared with the simulation results in terms of maximum water depths reached at 07:00 p.m. (12 December); by this time, operations for draining the flooded volume, which are not included in the simulation, had just started.

The model sensitivity to erodibility parameters was also analyzed. First, the simulation was repeated with a fixed critical shear stress ($\tau_c = 1$ Pa), and the erodibility coefficient was doubled (10 $cm^3 \cdot N^{-1} \cdot s^{-1}$), halved

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该区域堤顶高程并不均匀，某些低洼处更容易受到冲刷，从而形成多个小型溃口，并随时间逐渐扩大。

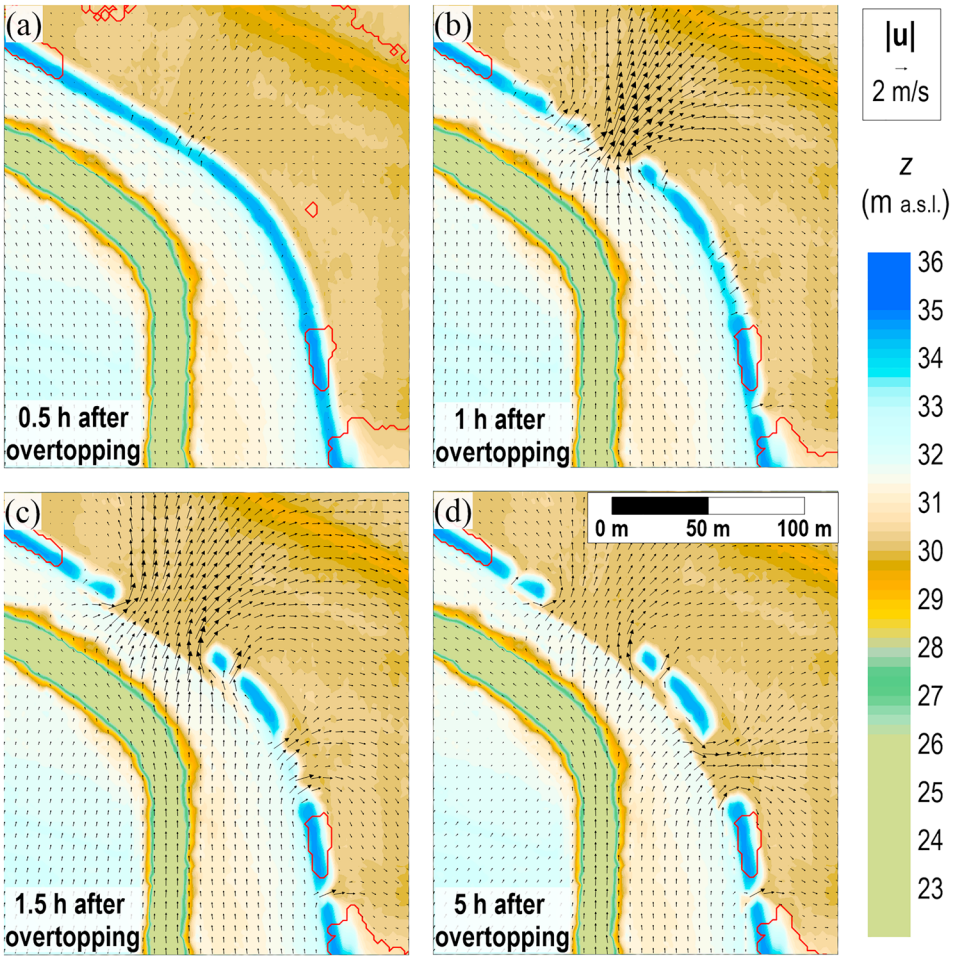

图7. 恩扎河堤防溃口。溃口区不同时刻模拟河床高程等值线图与流速矢量图：(a) 漫溢开始后0.5小时，(b) 1小时，(c) 1.5小时，(d) 5小时。为清晰起见，每九个矢量仅显示一个。红色等值线标识湿干交界面。

图7以不同时刻的河床高程等值线图（同时标示流速矢量）展示了溃口演化过程。初始阶段，侵蚀并未集中于某一受限区域（如前一测试案例所示），因为水流沿约270米长的堤顶漫溢，侵蚀过程沿此长度呈弥散分布。然而，该区域堤顶高程并不均匀，故某些低洼点更易遭受侵蚀，进而引发多个小型溃口的发育，且随时间推移逐渐扩大。其中最薄弱点位于堤防弯道下游侧，此处观测到最高流速。从漫溢开始至溃口达到最终规模耗时约3小时，此后2小时内仅有微小变化。最终总宽度约为150米，沿约250米长度分为五个区段，其中最大区段宽60米。实际上，现场仅观测到三个溃口；但考虑到边界条件、材料参数及地形高程等方面存在的不确定性，模型正确预测了多溃口开启，并较好地再现了整体过程。显然，任何仅依赖几何关系或河道与洪泛区之间内部联系来模拟溃口开启的模型，都难以捕捉此类复杂行为。

图8展示了实际总淹没区域，并与模拟所得12月12日19:00时刻的最大水深结果进行对比；此时，排水作业（模拟中未包含）刚刚开始。

还分析了模型对可蚀性参数的敏感性。首先，在临界剪应力（τc = 1 Pa）固定条件下重复模拟，并将可蚀性系数加倍（10 $cm^3 \cdot N^1 \cdot s^1$）、减半

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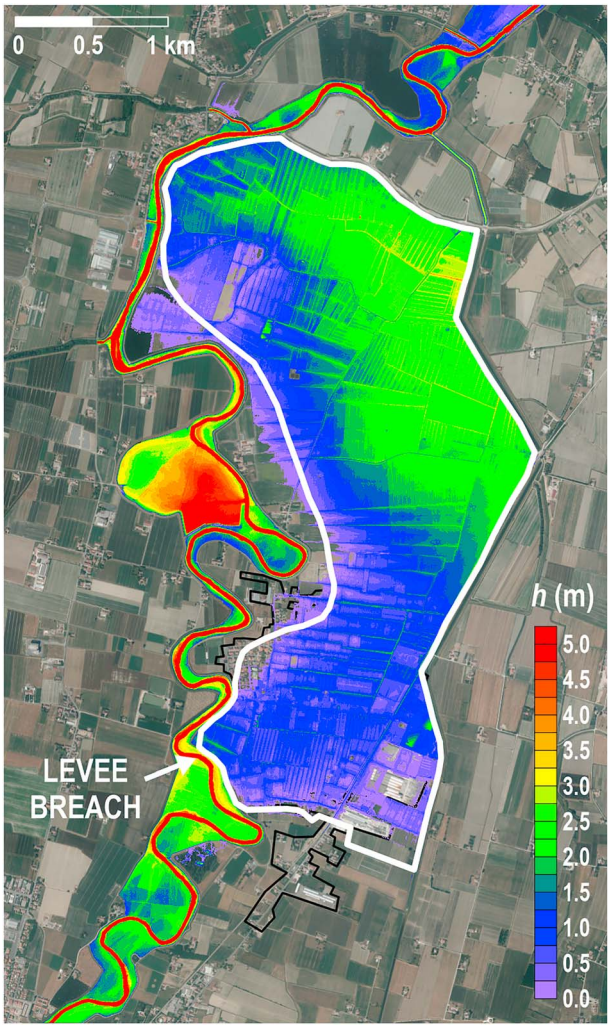

Figure 8. Levee breach on the Enza River. Contour map of the simulated maximum water depth (up to 07:00 p.m., 12 December) and actual total flooded area (white line).

(2.5 $cm^3 \cdot N^{-1} \cdot s^{-1}$), and further reduced to 1 $cm^3 \cdot N^{-1} \cdot s^{-1}$, in order to explore the whole range of variability of this parameter for the given material. The breach evolution for these simulations is compared in Figures 9a–9i. The erodibility coefficient has a significant impact on both the failure time and the breach evolution. When the value of k_d is reduced, the erosion process is slower and less pronounced. When k_d assumes the smallest value (1 $cm^3 \cdot N^{-1} \cdot s^{-1}$), the breach evolution takes 6 hr and is characterized by a generalized erosion along the whole overtopped length. Results of the simulation with $k_d = 2.5$ $cm^3 \cdot N^{-1} \cdot s^{-1}$ are similar to those of the reference case, even if the opening time slightly increases (4 hr). Surprisingly, while an increase in the value of k_d to 10 $cm^3 \cdot N^{-1} \cdot s^{-1}$ reduces the failure time to only 1.5 hr, the total width does not increase compared to that in the reference simulation. In fact, a single breach is generated: The most vulnerable portion of the levee is eroded very rapidly, and the consequent drop in the water level in the river stops the overtopping and erosion processes along the rest of the levee. A comparison of the breach outflow hydrographs is reported in Figure 9j for these simulations and shows that apart from the case with $k_d = 1$ $cm^3 \cdot N^{-1} \cdot s^{-1}$, which highly underestimates the outflow discharge (and the total flooded area), for the other values the peak discharge is underestimated by less than 10% compared to that in the reference simulation, and the peak is observed within ±0.5 hr. The differences in the total outflow volume are always below 10% compared to those in the reference simulation, except for the case with the lowest value of the erodibility coefficient, for which the volume difference is over 30%.

A sensitivity analysis to the critical shear stress was also performed, changing its value from 1 to 0.1 Pa, and to 10 Pa, maintaining $k_d = 5$ $cm^3 \cdot N^{-1} \cdot s^{-1}$, in order to evaluate its influence on the simulation results. The results confirm, similarly to the previous test cases, that the critical shear stress does not influence the breach evolution significantly, especially in the case of a reduced value assigned to this parameter. The adoption of the highest value, on the other hand, results in a slightly slower erosion process and reduced overflown volume of water (−15%), due to the fact that the predicted bed shear stresses and the critical value are of the same order of magnitude (10^1 Pa). Finally, the sensitivity to the critical angles for slope stability was also analyzed, but their influence on the simulation results is not evident for this test case (results not shown).

研究还采用了附录A所述的泥沙输运模型对此次洪水事件进行模拟

The flood event was also simulated using the sediment transport model described in Appendix A. The following parameters were assumed for the levee material: $d_{50} = 0.04$ mm, $d_{90}/d_{30} = 10$, and $\rho_s = 2{,}650$ kg/m^3. Sediment transport was allowed only in a wide region around the breach, not only to reduce the computational burden but also to better compare the two models. A single breach develops rapidly after levee overtopping: The failure time is less than 1 hr, and the final width is 65–75 m (a map of the breach site is reported in Figure 10). Actually, a similar behavior can be obtained from the erosion model if the erodibility coefficient is increased to 50 $cm^3 \cdot N^{-1} \cdot s^{-1}$, which, however, is no longer representative of the levee material. In fact, it must be stressed that a bedload transport model is not expected to describe the complex behavior of the erosion process in a levee built with cohesive material.

With regard to the simulation time, the erosion model takes 1.3 hr to simulate 2 days of physical time on a P100 Tesla® GPU, resulting in a ratio of physical to computational time equal to 37. Compared to than in an analogous simulation where the levee breach on the Enza River is modeled using a geometric approach (similarly to that in Vacondio et al., 2016), the computational overhead is negligible (3%). The sediment transport model takes 2.8 hr on the same device (physical/computational time = 17). The good

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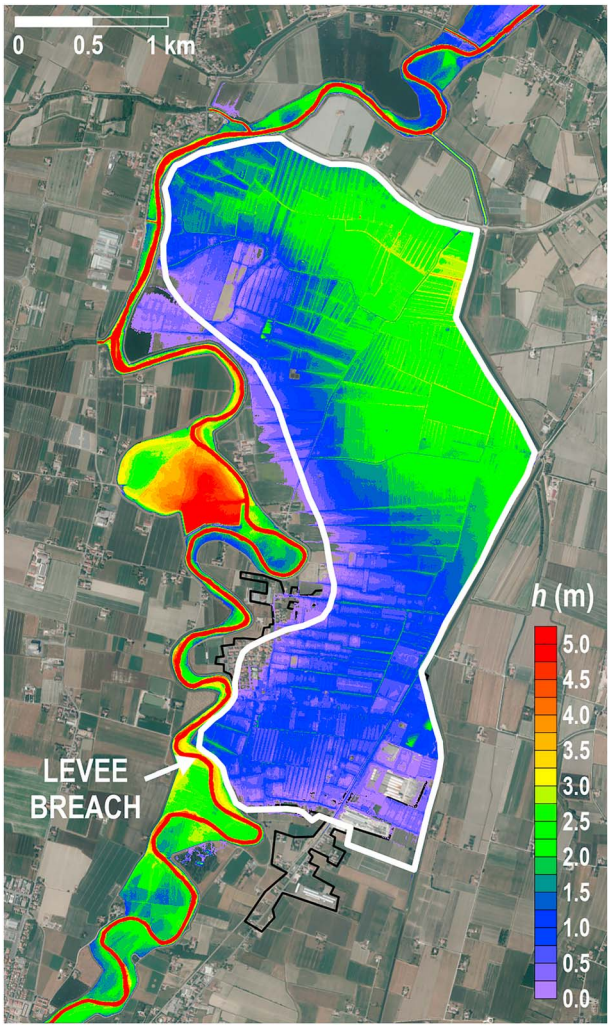

图8. 恩扎河堤防溃口。模拟所得最大水深等值线图（截至12月12日19:00）与实际总淹没区域（白色线）。

（2.5 $cm^3 \cdot N^1 \cdot s^1$），并进一步降至1 $cm^3 \cdot N^1 \cdot s^1$，以全面考察该参数在给定材料条件下的全部变化范围。这些模拟所对应的溃口演化过程见图9a–9i。可蚀性系数对溃决发生时间和溃口演化均具有显著影响。当kd值减小时，侵蚀过程变慢且程度减弱。当kd取最小值（1 $cm^3 \cdot N^1 \cdot s^1$）时，溃口演化历时6小时，且表现为整个漫顶段长度上的普遍侵蚀。kd = 2.5 $cm^3 \cdot N^1 \cdot s^1$时的模拟结果与基准案例相似，仅溃口开启时间略有延长（4小时）。出人意料的是，当kd增至10 $cm^3 \cdot N^1 \cdot s^1$时，溃决时间缩短至仅1.5小时，但总宽度并未较基准模拟有所增加。事实上，仅形成单一溃口：堤防最薄弱部位被极快侵蚀，随后河道水位迅速下降，从而终止了堤防其余部分的漫顶及侵蚀过程。图9j对比了上述模拟的溃口出流过程线，结果显示：除kd = 1 $cm^3 \cdot N^1 \cdot s^1$情形严重低估出流流量（及总淹没面积）外，其余情形的峰值流量较基准模拟的低估幅度均小于10%，且峰值出现时间偏差在±0.5小时以内。除可蚀性系数取最小值的情形（体积差超过30%）外，其余情形的总出流体积与基准模拟相比差异始终低于10%。

还开展了临界剪应力的敏感性分析，将其取值分别设为1 Pa、0.1 Pa和10 Pa，同时保持kd = 5 $cm^3 \cdot N^1 \cdot s^1$不变，以评估其对模拟结果的影响。结果表明，与前述测试案例类似，临界剪应力对溃口演化的影响并不显著，尤其当该参数取较小值时。而采用最大值时，则导致侵蚀过程略微减缓，溢流水量减少15%，原因是预测的床面剪应力与该临界值之间的关系所致。

量级相同（均为10^1 Pa）。最后，还分析了边坡稳定性临界角度的敏感性，但在此测试案例中，其对模拟结果的影响并不显著（结果未展示）。

研究还采用了附录A所述的泥沙输运模型对此次洪水事件进行模拟

该洪水事件亦采用附录A所述泥沙输运模型进行了模拟。堤防材料参数设定如下：d50 = 0.04 mm，d90/d30 = 10，ρs = 2650 kg/m^3。为减轻计算负担并便于两种模型间对比，泥沙输运仅允许发生在溃口周边较宽区域内。溃口在堤防漫顶后迅速形成：失事时间不足1小时，最终宽度为65–75米（溃口位置示意图见图10）。实际上，若将侵蚀模型中的可蚀性系数提升至50 $cm^3 \cdot N^1 \cdot s^1$，亦可获得类似行为；但该值已不再代表堤防材料的真实特性。须特别强调的是，推移质输运模型本就不适用于描述由黏性材料构筑的堤防所呈现的复杂侵蚀行为。

就计算耗时而言，侵蚀模型在P100 Tesla® GPU上模拟2天物理时间需1.3小时，即物理时间与计算时间之比为37。相较于采用几何方法（类似Vacondio等，2016年所用方法）模拟恩扎河堤防溃口的同类计算，其额外计算开销可忽略不计（仅3%）。同一设备上，泥沙输运模型耗时2.8小时（物理时间与计算时间之比为17）。良好的……

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冲刷系数太大，仅形成单一决口：堤防最薄弱的部分被迅速冲蚀，导致河道水位快速下降，从而中断了其余堤段的漫顶及冲刷过程

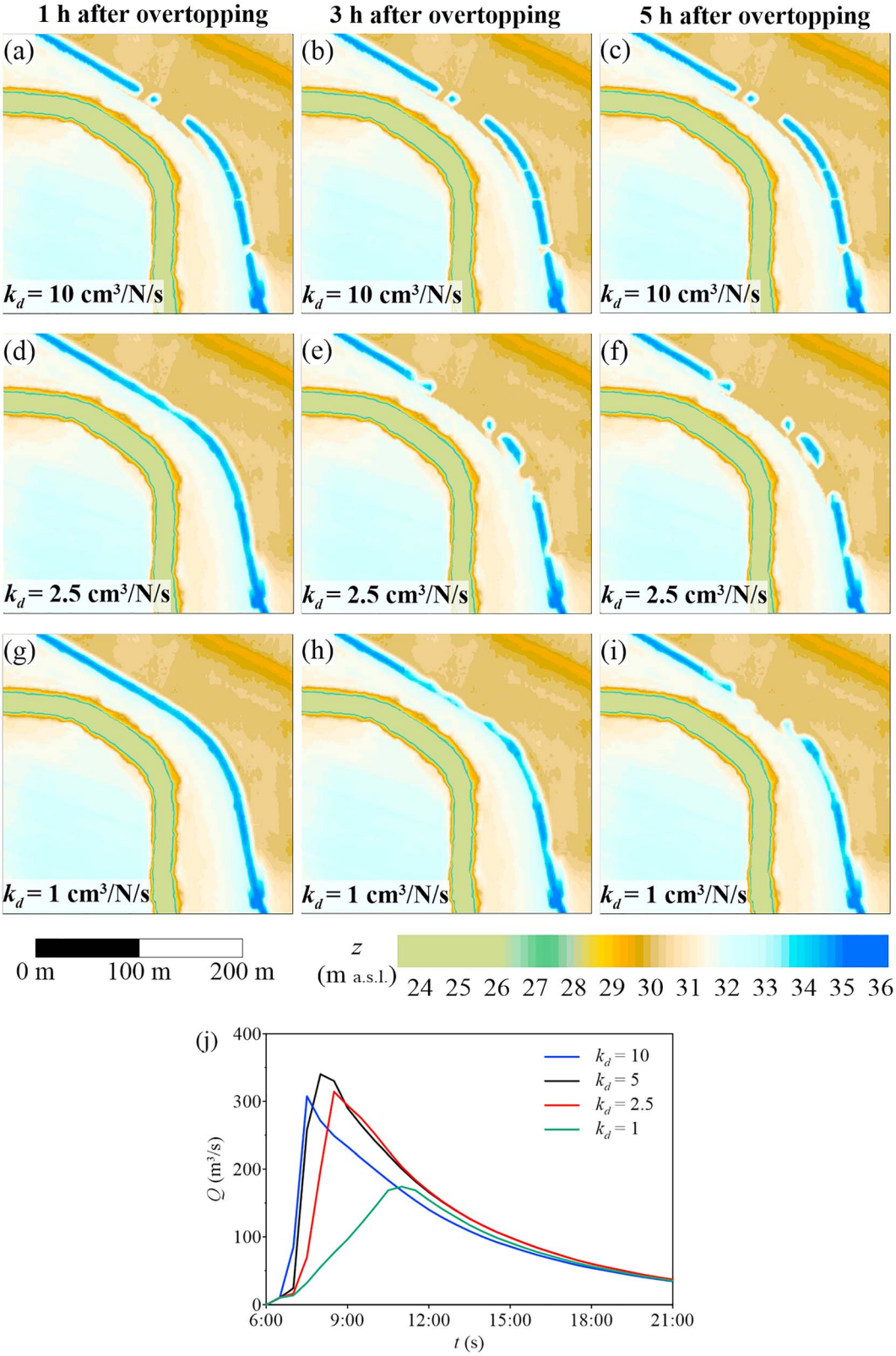

Figure 9. Levee breach on the Enza River: sensitivity analysis on the erodibility coefficient. Contour maps of the bottom elevation in the breach zone at selected times for the different simulations: (a–c) k_d = 10 cm^3·N^{-1}·s^{-1}; (d–f) k_d = 2.5 cm^3·N^{-1}·s^{-1}; (g–i) k_d = 1 cm^3·N^{-1}·s^{-1}. (j) Simulated outflow hydrographs from the breach for different values of the erodibility coefficient k_d (expressed in cm^3·N^{-1}·s^{-1}).

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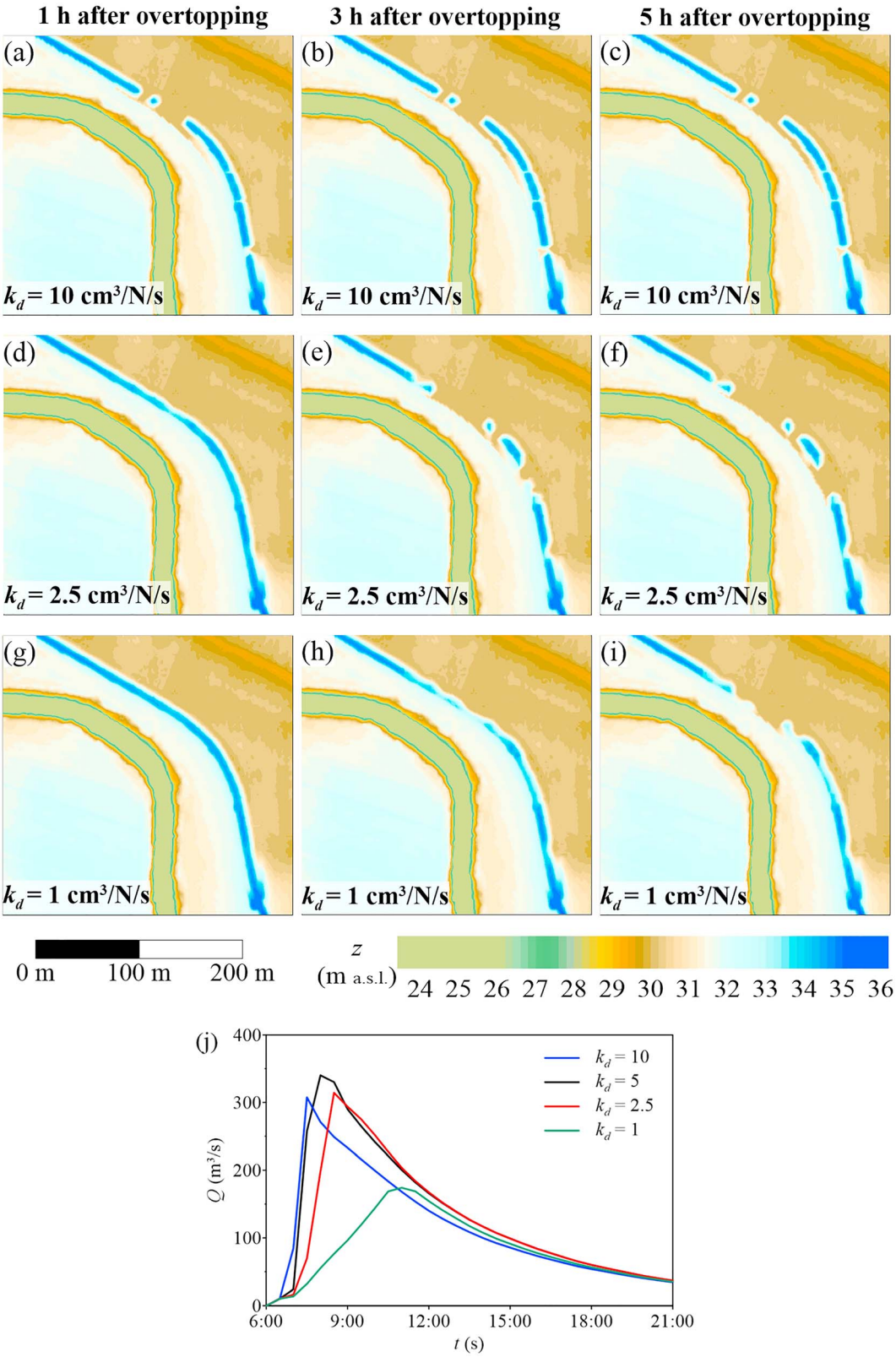

图9. 恩扎河堤防溃口：可蚀性系数敏感性分析。不同模拟工况下溃口区在选定时刻的河床高程等值线图： （a–c）kd = 10 cm³·N¹·s¹； （d–f）kd = 2.5 cm³·N¹·s¹； （g–i）kd = 1 cm³·N¹·s¹。 （j）不同可蚀性系数kd值（单位：cm³·N¹·s¹）对应的溃口出流过程线。

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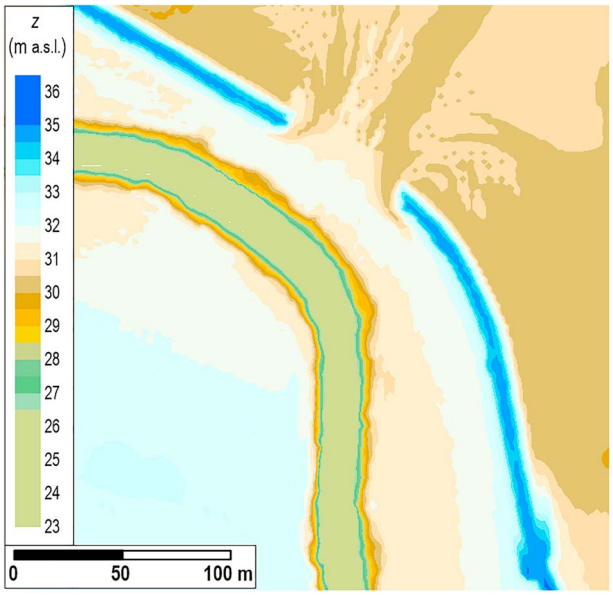

Figure 10. Levee breach on the Enza River. Contour map of the final bottom elevation in the breach zone obtained from the sediment transport model.

performance of GPU-accelerated models for high-resolution simulations is thus confirmed, and this makes the application of these models to complex real field test cases particularly convenient.

4. Discussion

The main aim of the present work is to introduce a tool able to automatically handle the possible opening of one or more levee breaches due to overtopping in a state-of-the-art 2D numerical code for flood propagation. The need to define the breach dimensions and position as input parameters is avoided by introducing an equation that predicts the bottom erosion wherever the local value of the bed shear stress exceeds a critical value (in the practice, this coincides with the occurrence of levee overtopping). The model was validated by simulating experimental test cases and one real flood event, and the numerical results were in good agreement with the measurements. In particular, the application to the historical test case of December 2017 on the Enza River, presented in section 3.4, shows that the proposed model can effectively simulate this kind of problems: The breach position is correctly identified, and the opening of multiple breaches can be captured; moreover, the predicted outflow hydrograph from the breach results in a well-reproduced flooded area.

Clearly, the proposed model does not capture the breach process in detail, because some of the many interrelated factors influencing the breach dynamics are neglected. The simplifications introduced in the model must be kept in mind (most of them are common to other simplified approaches). Among these, it is worth recalling that a hydrostatic pressure distribution is assumed in the depth-averaged model, while vertical accelerations and streamline curvatures are often nonnegligible at the breach site, especially in the case of headcut erosion in cohesive embankments. Moreover, other factors that may increase the bed shear stress or reduce the levee resistance to erosion, such as sediment concentration and infiltration processes, are not considered in the present model. Despite these simplifications, the final breach width and peak outflow discharge are predicted reasonably even in the case of cohesive embankments, as shown by results from Test #10 in section 3.2. Finally, the model assumes that the eroded material is washed away by the flow (i.e., sediment deposition is not simulated). This may slightly influence the breach configuration, especially in the case of noncohesive materials (see, for example, Figure 5), and the effect is more evident for small-scale experimental test cases. However, for real field simulations, this hypothesis does not probably have much impact on the overall process.

The influence of the levee material on the erosion process is taken into account by means of two model parameters, the erodibility coefficient and the critical shear stress. Although their definition is not straightforward, some indications from the literature allow identifying the plausible range of variability of their values. The sensitivity analyses performed for several test cases show that the critical shear stress does not influence the model results significantly, because the bed shear stresses largely exceed the critical value during overtopping. On the other hand, the erodibility coefficient must be defined more carefully, especially in the case of noncohesive material, for which its range of variability can be quite large. The analyses conducted in this study show that the breach width is affected by the value assigned to this parameter, while the outflow hydrograph is somewhat less influenced (as long as the value of k_d is representative of the material type), as can be noticed from results reported in section 3.3. Similar considerations are true for the case of a cohesive material (see results from section 3.4). This is encouraging, since the outflow hydrograph is the most relevant outcome of the model when the aim of the simulation is the prediction of the levee breach-induced flooding.

In all tests simulated in the present work, the embankment is considered homogeneous, meaning that a unique value of the erosion model parameters is adopted for the whole domain, but different values can easily be assigned to different segments of the levee; scour-dependent parameters could also be used in order to model stratified embankments. In general, however, real levees are heterogeneous, and data concerning the spatial distribution of the material type are often lacking, hence assigning a unique "average" value to

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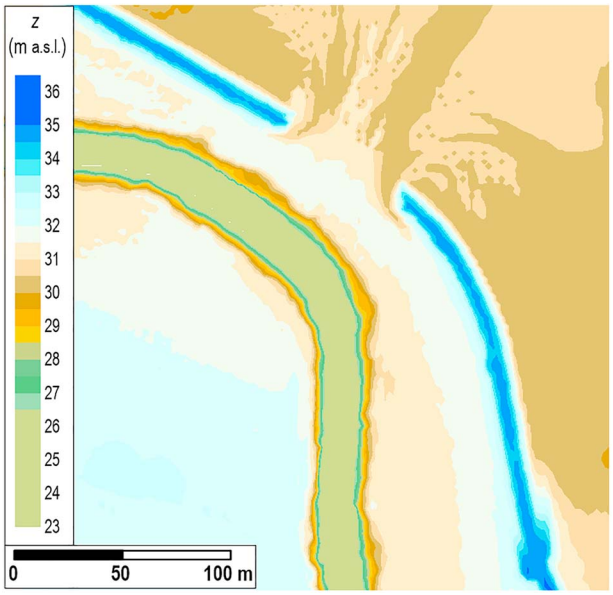

图10. 恩扎河堤防溃口：基于泥沙输运模型所得溃口区最终河床高程等值线图。

GPU加速模型在高分辨率模拟中的优异性能由此得到进一步验证，这使其在复杂真实野外案例中的应用尤为便捷。

4. 讨论

本研究的主要目标是开发一种工具，使其能在当前先进的二维洪水演进数值模型中自动处理因漫顶导致的一处或多处堤防溃口开启问题。通过引入一个侵蚀方程——当局部床面剪切应力超过临界值（实践中即对应堤防漫顶发生）时，该方程即可预测底部侵蚀——从而避免了将溃口尺寸与位置作为输入参数预先设定的需要。模型经由多个试验案例及一次真实洪水事件的模拟完成验证，数值结果与实测数据吻合良好。特别是第3.4节所介绍的2017年12月恩扎河历史试验案例表明，所提模型能有效模拟此类问题：溃口位置被准确识别，且可捕捉多处溃口同步开启现象；此外，溃口出流过程线的预测结果亦能较好复现实际淹没范围。

显然，所提出的模型无法精细刻画溃口过程，因其未能涵盖影响溃口形成的诸多相互关联因素。

水流垂向动力学过程被忽略。模型中引入的各类简化假设需予以注意（其中多数亦见于其他简化方法）。值得指出的是，该垂向平均模型假定压力沿水深呈静水分布，而溃口处垂向加速度与流线曲率往往不可忽略，尤其在黏性堤身发生溯源侵蚀时。此外，其他可能增大床面剪切应力或削弱堤防抗侵蚀能力的因素（如泥沙浓度与入渗过程）亦未纳入本模型考量。尽管存在上述简化，模型对溃口最终宽度与峰值出流量的预测结果仍较为合理，如第3.2节试验#10的结果所示。最后，模型假设被侵蚀物质完全被水流带走（即不模拟泥沙淤积）。此假设可能轻微影响溃口形态，尤其在非黏性材料情形下（参见图5），且在小尺度试验中表现更为明显。然而在真实野外模拟中，该假设对整体过程影响甚微。堤防材料对侵蚀过程的影响通过两个模型参数体现：可蚀性系数与临界剪切应力。尽管二者定义并非直接明了，但文献中已有若干参考依据，可帮助界定其合理取值范围。针对多个试验案例开展的敏感性分析表明，临界剪切应力对模型结果影响不大，因为在漫顶过程中床面剪切应力远超临界值；而可蚀性系数则需更审慎确定，尤其对于非黏性材料，其取值范围可能相当宽泛。本研究分析显示，溃口宽度受该参数取值影响显著，而出流过程线受其影响相对较小（只要kd值能合理表征材料类型），第3.3节结果即印证了这一点。对黏性材料情形亦可作类似判断（参见第3.4节结果）。这一现象令人鼓舞，因为当模拟目标为预测堤防溃口引发的洪水时，出流过程线正是模型最关键的输出结果。

在本研究模拟的所有试验中，堤防均被视为均质体，即全计算域采用统一的侵蚀模型参数值；但不同堤段亦可便捷地赋予不同参数值；此外，亦可采用与冲刷深度相关的参数以模拟成层堤防。然而总体而言，实际堤防多呈非均质性，且有关材料类型空间分布的数据往往缺失，因此常需赋予一个统一的平均值。

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the model parameters can be considered reasonable in the practice. A possible enhancement to this approach can be the inclusion of the presence of a grass cover, which protects the levee surface and delays the beginning of the erosion process, reducing the probability of levee failure (Mazzoleni et al., 2017). This effect can be achieved by increasing the critical shear stress, or by adjusting Manning's coefficient (as done by Viero et al., 2013), during the first stages of overtopping. Finally, a scour-dependent erodibility coefficient could be used to distinguish between the embankment material and the levee foundation material, when this cannot be assumed as non-erodible. These modifications to the proposed model will be considered in future works.

Results obtained with the erosion model were also compared with the predictions of a simple sediment transport model that integrates the Exner equation for bedload transport. The main drawbacks of this latter model were found to be not only the greater complexity and computational effort required but, more importantly, its inadequacy for simulating the erosion of cohesive levees. Moreover, even in the case of a noncohesive material, a sediment transport model may overestimate the embankment erosion process. In fact, experimental observations show that a fine sand is more resistant to erosion than a coarse sand, due to its "apparent cohesion" (Evangelista, 2015; Pickert et al., 2011), while bedload transport laws usually assume that the transport capacity increases when the size of particles decreases. More sophisticated sediment transport models could be employed, but only at the cost of introducing a larger number of model parameters and of further increasing the computational time. For these reasons, the employment of this kind of models does not seem justified for simulating the flooding triggered by levee breaches in very large domains.

Finally, the efficiency of the proposed model must be stressed. Previous work (e.g., Vacondio et al., 2014) already assessed that a speedup up to 2 orders of magnitude can be obtained using GPU-accelerated models (as PARFLOOD) instead of serial codes. The implementation of the proposed erosion model does not degrade the computational efficiency compared to the adoption of a simple geometric approach and requires a lower computational time than a sediment transport model.

5. Conclusions

In this paper, a 2D SWE code was coupled with an erosion model that allows simulating the opening of levee breaches generated by overtopping, without the need to set the breach position and dimensions as input parameters. The breach evolution was predicted correctly for all tests used for model validation, which concerned both cohesive and noncohesive materials. Therefore, the model can be particularly useful to create flood hazard maps and to support the design and verification of existing levee systems, also thanks to the high computational efficiency of the GPU implementation.

Appendix A: Sediment Transport Model

A simple bedload transport model was also developed for comparison purposes. The sediment transport model was taken from the literature, and the implementation of Juez et al. (2014) was adopted, also because it can be efficiently coded for GPUs. The model is based on the integration of the Exner equation, and hence, only bedload transport is considered; moreover, an approach where SWE and the morphodynamic equation are uncoupled is chosen (in order to reduce model complexity). For the hydrodynamic part, the model described in section 2.1 is adopted. Here only the main features of the sediment transport model are briefly recalled.

The bed evolution is described by the 2D Exner equation:

$$\frac{\partial z}{\partial t}+\frac{1}{1-p}\frac{\partial q_{s,x}}{\partial x}+\frac{1}{1-p}\frac{\partial q_{s,y}}{\partial y}=0, \tag{A1}$$

where $q_{s,x}$ and $q_{s,y}$ are the bed load discharges in the x and y directions, respectively, which can be expressed by any sediment transport formula; in this work, the Smart equation (Smart, 1984) is adopted:

$$\Phi=4\left(d_{90}/d_{30}\right)^{0.2}FS^{0.1}\theta^{0.5}\left(\theta-\theta_c^{SM}\right). \tag{A2}$$

In equation (A2), Φ is the dimensionless sediment discharge, computed according to the following expression:

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模型参数在工程实践中可视为合理。该方法一种可能的改进方向是考虑草皮护面的影响——草皮可保护堤防表面、延缓侵蚀起始时间，从而降低堤防失事概率（Mazzoleni等，2017）。这一效应可通过提高临界剪切应力或在漫顶初期调整曼宁系数（如Viero等，2013年所采用的方法）来实现。最后，还可采用冲刷深度相关的可蚀性系数，以区分堤身材料与堤基材料（当后者不能假定为不可蚀时）。上述对所提模型的改进将在后续研究中予以考虑。

采用侵蚀模型所得结果还与一个基于推移质输运Exner方程的简化泥沙输运模型的预测结果进行了对比。后者的突出缺陷不仅在于更高的复杂性和计算开销，更关键的是其难以准确模拟黏性堤防的侵蚀过程。此外，即便对于非黏性材料，泥沙输运模型仍可能高估堤防侵蚀速率。事实上，实验观测表明，细砂因具有表观黏聚力（Evangelista，2015；Pickert等，2011），其抗侵蚀能力反而强于粗砂；而推移质输运公式通常假设颗粒粒径越小，输运能力越强。虽可采用更复杂的泥沙输运模型，但势必引入更多模型参数并进一步增加计算耗时。正因如此，在大范围区域模拟由堤防溃口引发的洪水时，采用此类模型似乎并无必要。

最后，必须强调所提出模型的高效性。先前的研究（例如Vacondio等，2014年）已证实，采用GPU加速模型（如PARFLOOD）相比串行代码可获得高达两个数量级的加速比。所提出的侵蚀模型的实现并未降低计算效率，其效率与采用简单几何方法相当，且所需计算时间低于泥沙输运模型。

5. 结论

本文将一个二维浅水方程（SWE）代码与一个侵蚀模型耦合，该模型可在不预先设定溃口位置和尺寸作为输入参数的情况下，模拟由漫顶引发的堤防溃口开启过程。在所有用于模型验证的试验中，溃口演化过程均被准确预测，这些试验涵盖了黏性与非黏性两类材料。因此，该模型特别适用于绘制洪水风险图，并支持现有堤防系统的设计与校核，此外，其GPU实现所具有的高计算效率亦为此提供了有力支撑。

附录A：泥沙输运模型

本文还构建了一个简单的推移质输运模型用于对比分析。该泥沙输运模型取自文献，采用了Juez等（2014）的实现方案，因其亦适用于GPU高效编程。该模型基于埃克纳方程的积分，故仅考虑推移质输运；此外，选取了浅水方程（SWE）与地貌动力学方程解耦的方法（以降低模型复杂度）。水动力部分采用第2.1节所述模型。此处仅简要回顾泥沙输运模型的主要特征。

床面演变由二维埃克纳方程描述：

$$\frac{\partial z}{\partial t} + \frac{1}{1-p}\frac{\partial q_{s,x}}{\partial x} + \frac{1}{1-p}\frac{\partial q_{s,y}}{\partial y} = 0; \tag{A1}$$

其中qs,x与qs,y分别为x方向与y方向的推移质输沙率，可采用任意泥沙输运公式表达；本研究采用Smart公式（Smart，1984）：

$$\Phi = 4\,(d90/d30)^{0.2} S^{0.6} \left(\theta - \theta_{SMc}\right)^{1.5}. \tag{A2}$$

在方程（A2）中，Φ为无量纲泥沙输沙率，按以下表达式计算：

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$$\Phi = \frac{|q_s|}{\sqrt{g(s-1)d_m^3}}, \tag{A3}$$

where $s = \rho_s/\rho$ is the ratio of the material density ρ_s to the water density ρ and d_m is the mean sediment diameter. Moreover, equation (A2) contains the Shields parameter θ (i.e., the dimensionless shear stress τ), defined as follows:

$$\theta = \frac{\tau}{g(\rho_s-\rho)d_m}, \tag{A4}$$

and θ_c^{SM} is its critical value according to Smart (1984); d_{90} and d_{30} are the grain sizes at which 90% and 30%, respectively, by weight of material is finer, F is the Froude number, and S is the friction slope. Equation (A2) is applied only when $\theta > \theta_c^{SM}$ (otherwise, $\Phi = 0$).

The bed elevation is updated as follows:

$$z_{i,j}^{n+1} = z_{i,j}^{n} - \frac{\Delta t}{\Delta x}\left(q^*_{s,i+\frac{1}{2},j} - q^*_{s,i-\frac{1}{2},j}\right) - \frac{\Delta t}{\Delta y}\left(q^*_{s,i,j+\frac{1}{2}} - q^*_{s,i,j-\frac{1}{2}}\right), \tag{A5}$$

where q^*_s represents the "sediment" flux at each intercell. Only the computation at the intercell between cells (i, j) and $(i + 1, j)$ is reported in the following (for the other three fluxes, similar expressions can be used). Flux is calculated as

$$q^*_{s,i+\frac{1}{2},j} = \begin{cases} \frac{1}{1-p} q_{s,x,i,j} & \text{if } \widetilde{\lambda}_s > 0, \\ \frac{1}{1-p} q_{s,x,i+1,j} & \text{if } \widetilde{\lambda}_s \le 0. \end{cases} \tag{A6}$$

The "sediment" celerity $\widetilde{\lambda}_s$ is estimated as

$$\widetilde{\lambda}_s = \frac{\delta q_{s,x,}}{\delta z}, \tag{A7}$$

where $\delta q_{s,x} = q_{s,x,i+1,j} - q_{s,x,i,j}$, and $\delta z = z_{i+1,j} - z_{i,j}$.

The bank failure algorithm presented in section 2.3 is also added to the model.

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$$\Phi = \frac{|q_s|}{\sqrt{g\,s-1\,()\,d\bar{z}n}}, \tag{A3}$$

其中s = ρs/ρ为固体材料密度ρs与水体密度ρ之比，dm为泥沙平均粒径。此外，方程（A2）中包含谢尔德参数θ（即无量纲剪切应力τ），其定义如下：

$$\theta = \frac{\tau}{g\,\rho s-\rho\,()\,dm}, \tag{A4}$$

及θSMc为Smart（1984）提出的临界值；d90和d30分别为按质量计有90%和30%颗粒更细的粒径；F为弗劳德数；S为摩阻坡度。方程（A2）仅在θ > θSMc时应用（否则Φ = 0）。

床面高程按如下方式更新：

$$z_{n+1}^{i;j} = z_n^{i;j} - \frac{\Delta t}{\Delta x}\left(q^*_{s,i+\frac{1}{2},j} - q^*_{s,i-\frac{1}{2},j}\right) - \frac{\Delta t}{\Delta y}\left(q^*_{s;i;j+1\bar{2}} - q^*_{s,i,j-\frac{1}{2}}\right), \tag{A5}$$

其中q*s表示各网格交界面处的泥沙通量。下文仅给出网格(i, j)与(i + 1, j)之间交界面处的计算结果（其余三个通量可采用类似表达式）。通量按如下方式计算：

$$q^*_{s;i+\bar{2};j} = \begin{cases} \frac{1}{1-p} q_{s;x;i;j} & (\text{若} e\lambda s > 0) \\ \frac{1}{1-p} q_{s;x;i\flat 1;j} & (\text{当} e\lambda s \le 0\text{时}) \end{cases}. \tag{A6}$$

泥沙传播速度eλs按如下方式估算：

$$\widetilde{\lambda}_s = \frac{\delta q_{s,x,}}{\delta z}, \tag{A7}$$

其中δqs,x = qs,x,i+1,j − qs,x,i,j，δz = zi+1,j − zi,j。

第2.3节提出的岸坡失稳算法亦已加入本模型。

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